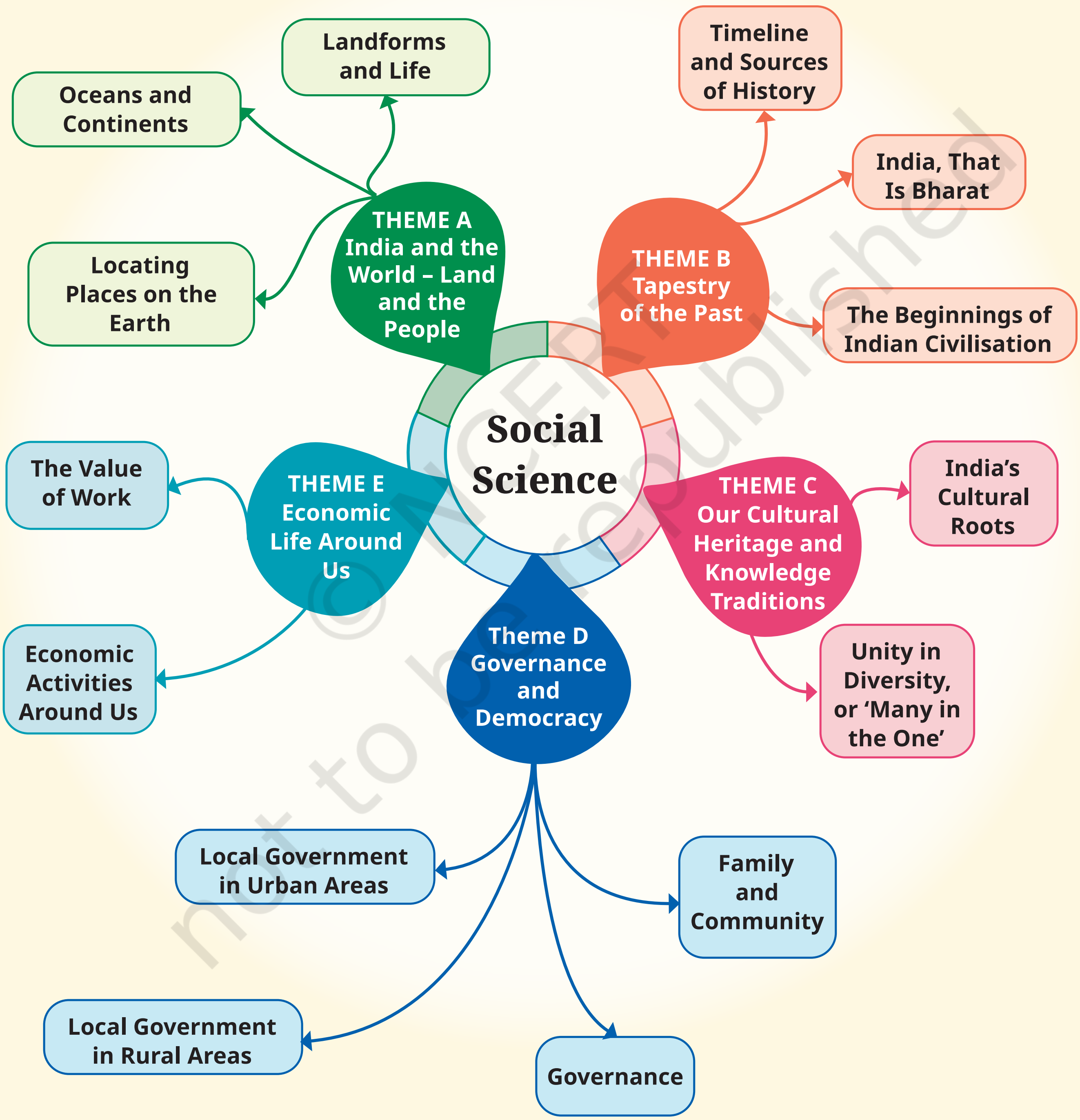
Introduction
Why Social Science?



LET'S EXPLORE

→ Observe the picture above. What do you notice?
  - Where does the water in the lake come from?
  - Who made the road and why?
  - What could be the activities of people living in the small house? What could be their history? Their future?

→ Write down your answers and discuss them with your classmates.

→ Now, looking at the picture on the facing page, what questions come to your mind? Write them down.

→ How do you propose to find answers to the questions related to these two images?

How are our questions above relevant to Social Science?

We live in the 21st century (if you do not know what this figure really means, you will soon learn about it). Everyone agrees that it is a particularly challenging time for humanity. On

Reprint 2025-26

the one hand, there is rapid progress in technology, which is changing our lives in many ways. On the other hand, the world is witnessing multiple wars, armed conflicts and rising social tensions, and our planet's natural environment is under great stress. We live in an age of great possibilities but also great challenges.

The world over, more and more people wonder, "How do we solve the problems facing humanity? How can our societies learn to live in peace and harmony? How can we protect this beautiful Earth which we all share — and protect it not only for ourselves but also for all the species that live on it?"

These fundamental questions are simple, but the answers are not. They cannot be simple, because human societies are very diverse and complex. If we wish to find answers to such questions and help build a better future, we first need to understand our world, and human societies in particular. That is what Social Science is all about.

You may wonder whether this is a 'science' like, say, physics or chemistry. It is not. The discipline does use scientific

Reprint 2025-26

methods wherever possible (you will see a few examples in this textbook), but its focus — human society — is, again, too diverse to allow the kind of set procedures and fixed results the sciences come up with.

Social Science has many subdisciplines: geography, history, political science, economics, sociology, anthropology, archaeology, psychology and a few more. You need not feel intimidated by all these terms! While you will study some of these subdisciplines in the Secondary Stage, in the Middle Stage we have avoided this classification. Instead, we have opted for five broad themes. Let us briefly look at them.

Theme A – India and the World: Land and the People

This first theme includes the basics of the geographical world around us — some of the main features of our planet and the way to represent them on a map. Why is this theme important, when today we can get excellent maps on a mobile phone? Because it deals with much more than maps. It also asks how geographical features — oceans, mountains, rivers, etc. — have shaped entire civilisations throughout their histories. It is also, in India's case, about how its natural setting has contributed to giving this ancient civilisation a unique identity.

Theme B – Tapestry of the Past

A tapestry is a large piece of canvas-like cloth usually kept as a wall hanging, with pictures and designs on it — sometimes a historical narrative. Our tapestry is where we will be painting scenes from the past, beginning with India's past. But why should we be at all concerned with the past? Because it is the key to understanding the present, and the chapters in this theme will often make this connection clear. The past is a major source for our identities — it helps us understand who we are and where we come from.

Reprint 2025-26

The past is still with us, in other words. And since history is unfortunately not all about happy developments, it is useful to understand where people, governments or rulers went wrong, and why. Only then can we hope to avoid repeating those errors.

Theme C – Our Cultural Heritage and Knowledge Traditions

It has often been said that India has a rich and ancient culture. True, but what are its main characteristics? Its guiding principles? How has it manifested itself in India's history? And how can it help us to deal with issues of our times? These are some of the questions that this theme is exploring, with the objective that every student should understand some of the cultural foundations of our civilisation and learn to appreciate their value.

Theme D – Governance and Democracy

Citizens of any country should know how their political system functions. India, as the world's largest democracy, has an elaborate system working at different levels. What are its chief characteristics and components? How do the citizens participate in the overall governance? What are their rights and also their duties or dharma? Are there different systems in other countries, and, if so, of what type? How are different countries supposed to interact? By studying this theme, we can become more responsible citizens, understand how the organs of the government function, and learn to have a say in the policies that affect us all, whether locally or nationally.

Theme E – Economic Life Around Us

No family can be happy without the essentials of daily living — at least food, clothing, shelter, access to water in a first stage; in a second, livelihood for adults and access to education for the younger ones. Similarly, no country can

Reprint 2025-26

develop harmoniously without a sound economy. But how does an economy work, especially in a huge country like India? What exactly is money? Where does it come from? How can it be increased? What economic activities can people engage themselves in? How are natural and human resources best managed? This theme will lay down some of the important concepts and practices that will enable us to answer these questions.

You will notice that there are many questions in the preceding paragraphs. This is as it should be — Social Science is also about the art of asking the right questions. Only then can we start looking for the right answers. This also explains the presence of 'Big Questions' at the start of each chapter in this book.

You may also be intrigued to find a game of chess and some ancient Tamil poetry in chapters that apparently deal with geography; a discussion on the uses of the sari in a chapter on cultural heritage; the concept of *sevā* and the mention of festivals in chapters focusing on economics. This is deliberate. We believe in bringing elements from diverse fields together (you will learn later that this is called 'multidisciplinarity'). This enriches our perspective. Indeed, life constantly mixes numerous elements together, so why should we not?

By now, it should be clear that although Social Science makes constant use of the past, it seeks to make sense of the present so as to help us prepare a better future. It is an exploration and an adventure.

Reprint 2025-26

Locating Places on the Earth

CHAPTER 1

> *The globe of the Earth stands in space, made up of water, earth, fire and air and is spherical. ... It is surrounded by all creatures, terrestrial as well as aquatic.*
>
> — *Āryabhaṭa (about 500 CE)*

The Big Questions?

1. *What is a map and how do we use it? What are its main components?*
2. *What are coordinates? How can latitude and longitude be used to mark any location on the Earth?*
3. *How are local time and standard time related to longitude?*

0681CH01

Imagine that you are visiting a city for the first time. How would you find the places you want to visit? You might ask a local person for help, or you might look at a map of the city. In previous grades, you learnt a little about maps, and in this chapter, we will study them in more detail.

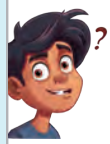
Let us play a game. Examine the map of this small city (Fig. 1.1). Imagine that you just got off a train at the railway station, and you want to visit the bank marked on the map. Which way would you go? Are there other possible ways? Can you locate the public garden, the school and the museum? If you want to proceed from the bank to the market, which way will you go? This is where a map comes in handy!

Fig. 1.1. A map of an imaginary small city.

A map is like a treasure guide; it shows you where things are and how to get to them. Notice the four arrows in the top right corner of the map; we will soon see how they point to some specific directions and make maps even more helpful.

Reprint 2025-26

LET'S EXPLORE



→ On the map in Fig. 1.1 given on page 8 —

1. Mark the hospital.
2. What is the meaning of the blue-coloured areas?
3. Which is farther away from the railway station — the school, the Nagar Panchayat or the public garden?

→ As a class activity, form groups of three or four students each. Let each group try to draw a map of your school and some of the streets or roads that lead to it, and a few neighbouring buildings. At the end, compare all the maps and discuss.

A Map and Its Components

From this simple example, we understand that a map is a representation, or a drawing, of some area — it may be a small area (a village, a town, etc.), a bigger area (say, your district or state), or a very large area like India or even the whole world. In a map, you look at the surface as if you are viewing it from the top.

An **atlas** is a book or collection of maps.

As you will discover, there are several kinds of maps —

- **physical maps,** which mainly show some natural features such as mountains, oceans and rivers (see an example in Fig. 5.2 in this textbook)
- **political maps,** which show details of countries or states, boundaries, cities, etc. (for instance, a map of India with all its States, Union Territories and their capitals)
- **thematic maps,** with a specific kind of information (examples include Fig. 6.3 and Fig. 8.1 in this textbook).

In addition, there are three important components of maps—**distance, direction** and **symbols**. You have already

Reprint 2025-26

experienced the first two while navigating the map in Fig. 1.1. Let us now define them more precisely.

Have you ever wondered how a huge place can fit on a small piece of paper? It is all thanks to the map's **scale**. Let us go back to our map of a small city (Fig. 1.1). Each centimetre on the map, as printed here, represents a certain distance on the ground — let us suppose it is 500 metres; we say that the scale is 1 cm = 500 m. Now, turn to the map of India in Fig. 5.2 in Chapter 5 of this textbook. The scale is represented in the bottom left corner by a ruler with '500' written above its length and 'km' on the side. It simply means that this ruler, which measures 2.5 cm in the printed map, corresponds to 500 kilometres on the ground.

So, the actual **distance** between two points represented on the map depends on the scale that the map is using.

LET'S EXPLORE

→ Draw a simple map of a school's playground. Let us assume it is a rectangle, 40 m in length and 30 m in width. Draw it precisely with your ruler on a scale of 1 cm = 10 m.

→ Now measure the diagonal of the rectangle. How many centimetres do you get? Using the scale, calculate the real length of the playground's diagonal, in metres.

Let us return to the four arrows at the top right of the small city's map. They point to four **directions**, which are north, at the top, and, moving clockwise, east, south and west. These are called the **cardinal directions**, also **cardinal points**. Other than these, intermediate directions are also used — northeast (NE), southeast (SE), southwest (SW) and northwest (NW). Most maps simply have an arrow marked with the letter 'N', which points to the north direction.

LET'S EXPLORE

→ Consider the map of the small city again. Identify the correct and incorrect statements in the list below:

1. The market is north of the hospital.
2. The museum is southeast of the bank.
3. The railway station is northwest of the hospital.
4. The lake is northwest of the apartment blocks.

→ Taking your school as the starting point, do you know approximately in which cardinal direction your home is located? Discuss with your teacher and your parents.

Symbols are another important component of maps. Our map has small drawings of actual buildings and a few other elements, but there would not be enough space on the map of a large city or a country to draw them all. Instead, a symbol is used to represent these features — symbols for different kinds of buildings (for instance a railway station, a school, a post office), for roads and railway lines, and for natural elements such as a river, a pond or a forest. In that way, numerous details can be shown in the limited space available on a map.

To make maps more easily understood by a variety of users, map makers use specific symbols. Different countries use different sets of symbols. The Survey of India, a government body, has fixed a set of symbols for maps of India (or parts of India). A small selection of them is shown in Fig. 1.2 on page 12.

LET'S EXPLORE

Draw a rough map of your locality or your village, including your home, school and a few other important landmarks. Show the cardinal directions and use a few of the symbols shown in Fig. 1.2 on page 12 to mark some important features.

Reprint 2025-26

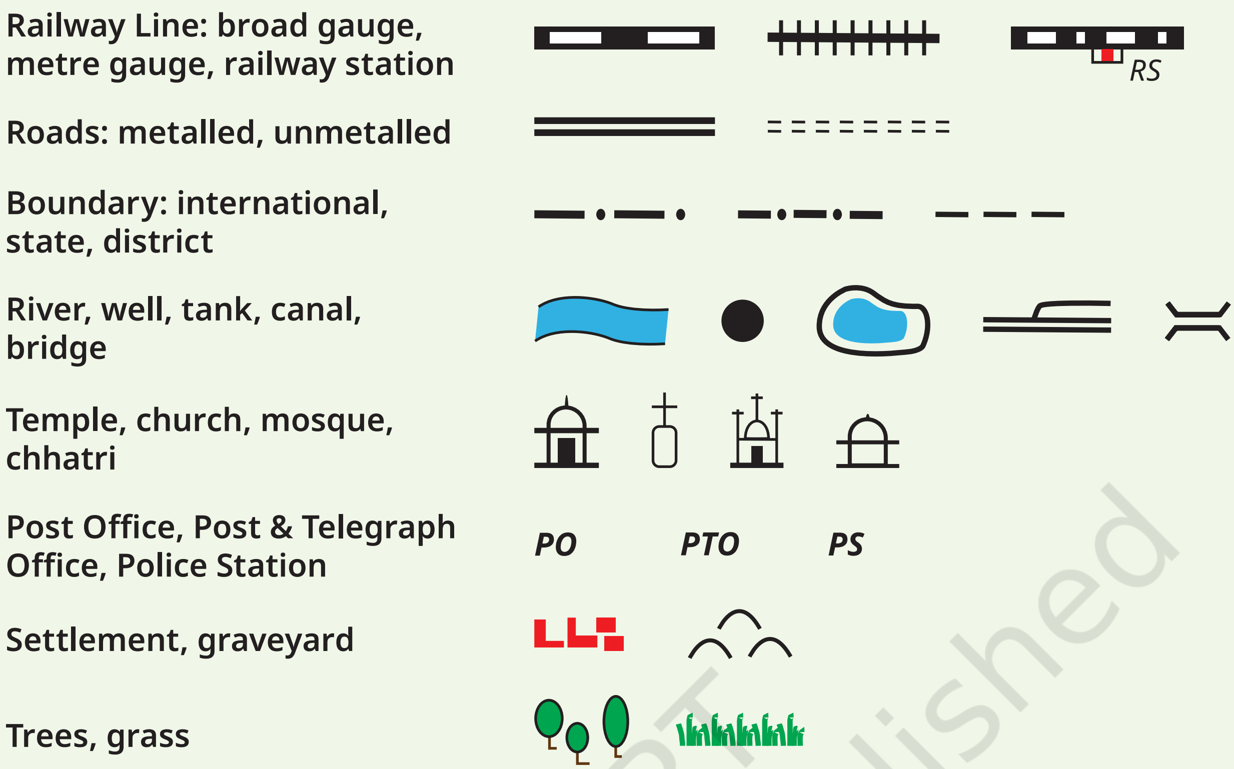


Fig. 1.2. A selection of symbols commonly used in maps.

Mapping the Earth

Mapping the Earth is a little more difficult because our planet is not a flat surface. It nearly has the shape of a sphere. (We say 'nearly' because it is not a perfect sphere, but is slightly flattened at the poles. However, in practice, we will consider it to be spherical.) Representing a sphere accurately on a flat sheet of paper is not possible. To understand why, peel an orange in such a way that you have just three or four large pieces of the skin; then try and flatten them on a table — you will see that you cannot do it without tearing them at the edges.

Now, consider a **globe**, which is a sphere on which a map is drawn. This may be a map of the Earth, the Moon, the planet Mars, the stars and constellations in the sky, etc. The physical object, like the one shown in the drawing on the next page, is a sphere that is generally made of metal, plastic or cardboard.

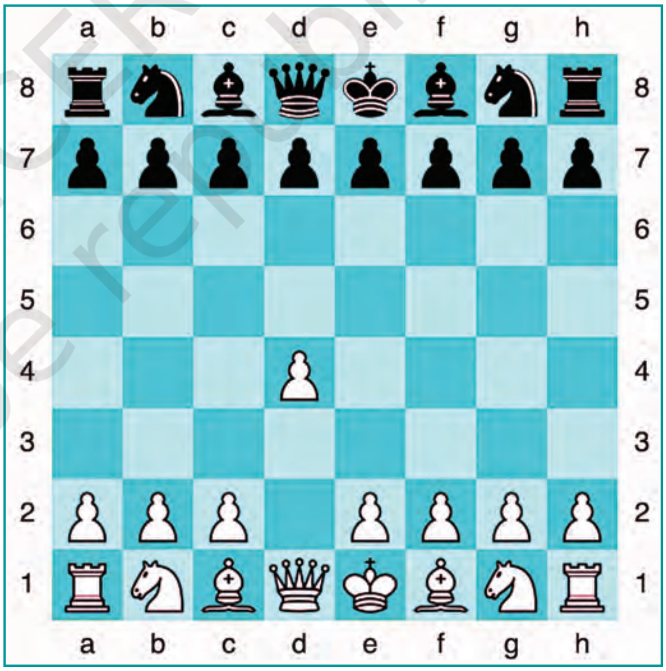
Reprint 2025-26

Here we will study the globe representing the Earth's geography. Because the globe and the Earth have the same spherical shape, a globe will better represent the geography of the Earth than a flat map.

Let us now explore some of its features.

a) Understanding coordinates

Imagine a big market in a city or town, with neat rows of shops, all the same size. You want to meet a friend at a stationery shop inside the market. But your friend does not know where the shop is. So you would give directions like, "Meet me at 6 pm at the 7th shop in the 5th row from the entrance." This will allow your friend to precisely determine your location.



Now, consider a chessboard. To record moves by advanced players, letters are placed alongside the main pieces (from 'a' to 'h', see the image) and numbers (from 1 to 8) in between the two sides. This simple system allows players to mark each square and record every move. Here, the white side has just opened the game by moving the queen's pawn two squares forward (a very common opening). So, the pawn is said to have moved from d2 to d4.

LET'S EXPLORE

Using the same terms, write down your move if you play black and respond with the same move.

Reprint 2025-26

The system used in these two examples may be called a system of **coordinates**. Thanks to their two coordinates, the stationery shop as well as the chess square on the chessboard can be precisely determined.

A similar system of coordinates is used in the world of maps to determine the location of any place on a map. Let us see how this system works.

b) Latitudes

Let us return to the globe. It is easy to identify the North Pole and the South Pole on it. Rotate the globe; while it rotates, the fixed points at the top and bottom are the two poles. Halfway between them is the Equator; note the circle marking it (see Fig. 1.3).

Imagine that you stand on the **Equator** and travel towards one of the poles; your distance from the Equator increases. **Latitude** measures this distance from the Equator. At any point of this travel, you can draw an imaginary line that runs east and west, parallel to the Equator. Such a line is called a **parallel of latitude** and it draws a circle around the Earth. Again, it is easy to note on the globe that the largest circle is the Equator, while the circles marked by the parallels of latitude grow smaller as we move northward or southward (Fig. 1.3).

Latitudes are expressed in **degrees**; by convention, the Equator is latitude 0° (zero degree), while the latitudes of the two poles are 90° North and 90° South respectively; this is noted 90°N and 90°S.

There is a connection between latitude and climate. Around the Equator, the climate is generally hot (it is also called 'torrid'). As you travel away from the Equator towards one of the two poles (in other words, as your latitude increases), the climate becomes more moderate (or 'temperate'). And closer to the North or South Pole, the climate grows colder (or 'frigid'). You will learn in Science why this is so, and

Reprint 2025-26

also why we experience different seasons in the course of a year.

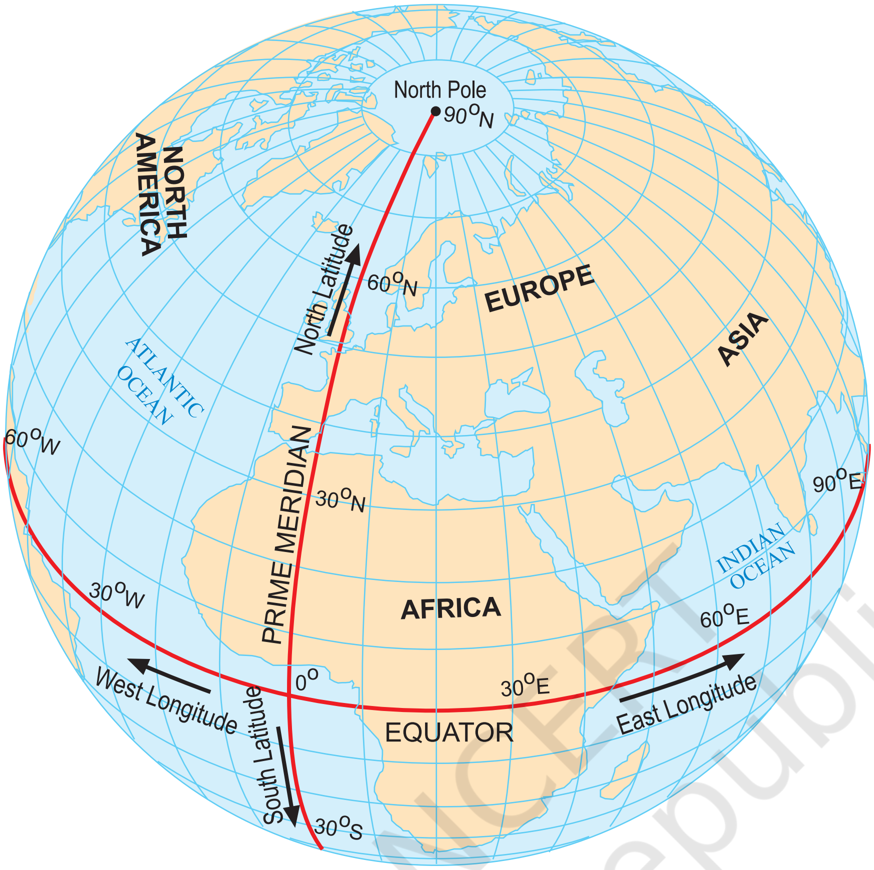


Fig. 1.3. This globe shows both parallels of latitude and meridians of longitudes

c) Longitudes

Imagine now that you travel from the North Pole to the South Pole by the shortest possible line. Observe the globe: you will see that instead of passing through Europe and Africa, you could just as well pass through Asia — the distance would be the same. These lines are called **meridians of longitude** (Fig. 1.3). They are all half-circles running from one pole to the other.

You will also learn in Science that the Earth spins on its axis. To put it simply, let's place a desk lamp a little away from our globe, focused on it, and imagine that this is the Sun illuminating the Earth. Slowly rotating the globe eastward, we can note that for some places on the Earth it is morning, while for others it is mid-day, evening or night — when it's breakfast time in one country, it's lunchtime in another and in a third country people are fast asleep! That is why by measuring the longitude of a place, we will also be measuring the time at that place. Let us see how.

To measure longitudes, we need to define a reference point called the **Prime Meridian** (Fig. 1.3 on page 15). It is also called Greenwich Meridian because, in the year 1884, some nations decided that the meridian passing through Greenwich, an area of London in England, would become the international standard for the Prime Meridian. It is marked as 0° longitude.

Just as latitude is a measure of the distance from the Equator if you travel towards one of the poles, **longitude** is a measure of the distance from the Prime Meridian if you travel along the Equator. Longitude, too, is measured in degrees. Westward or eastward, it increases in value from 0° to 180°, with the letter 'W' or 'E' added. For instance, using round figures, New York's longitude is 74°W, while Delhi's is 77°E and Tokyo's is 140°E.

DON'T MISS OUT

As you can see on the globe of meridians of longitudes, 180°W and 180°E are the same longitude; so this longitude is noted 180°, omitting the letter W or E.

Latitude and longitude together are the two **coordinates** of a place. With them, you are now able to locate any place on Earth! You can now understand a statement such as "Delhi lies at 29°N latitude and 77°E longitude" (these values are rounded off, not exact).

Fig. 1.3 on page 15 shows the parallels of latitude and the meridians of longitude together on the globe as blue lines.

Reprint 2025-26

All these lines together constitute a **grid** for the globe; they are also called grid lines.

LET'S EXPLORE

If the globe or atlas in your class has well-marked latitudes and longitudes, try to note down approximate values for the latitude and longitude of (1) Mumbai, (2) Kolkata, (3) Singapore, (4) Paris.

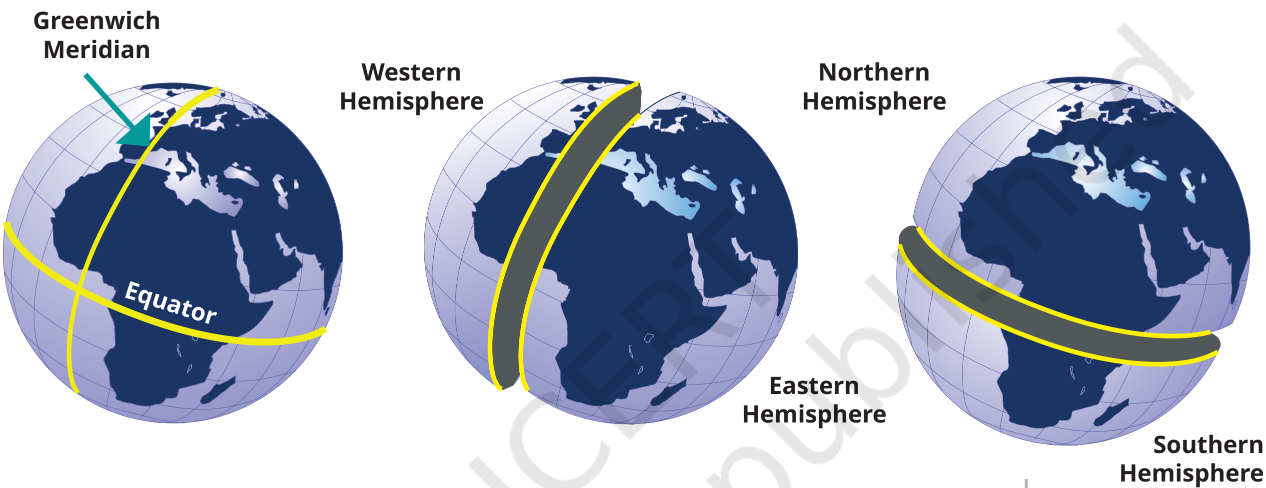


Fig. 1.4. This sketch shows how the Prime Meridian divides the Earth into the Western and Eastern Hemispheres, while the Equator divides it into the Northern and Southern Hemispheres.

DON'T MISS OUT

The Greenwich Meridian is not the first prime meridian. There were others in the past. In fact, many centuries before Europe, India had a prime meridian of its own! (Fig. 1.5) It was called *madhya rekhā* (or 'middle line') and passed through the city of Ujjayinī (today Ujjain), which was a reputed centre for astronomy over many centuries. Varāhamihira, a famous astronomer, lived and worked there some 1,500 years ago.

Indian astronomers were aware of concepts of latitude and longitude, including the need for a zero or prime meridian. The Ujjayinī meridian became a reference for calculations in all Indian astronomical texts.

Reprint 2025-26

The map shows a few ancient Indian cities close to the Ujjayinī meridian. Some are very close to it, while others are a little away. That is because measuring longitude required accurate timekeeping, which was not as precise then as it is today.

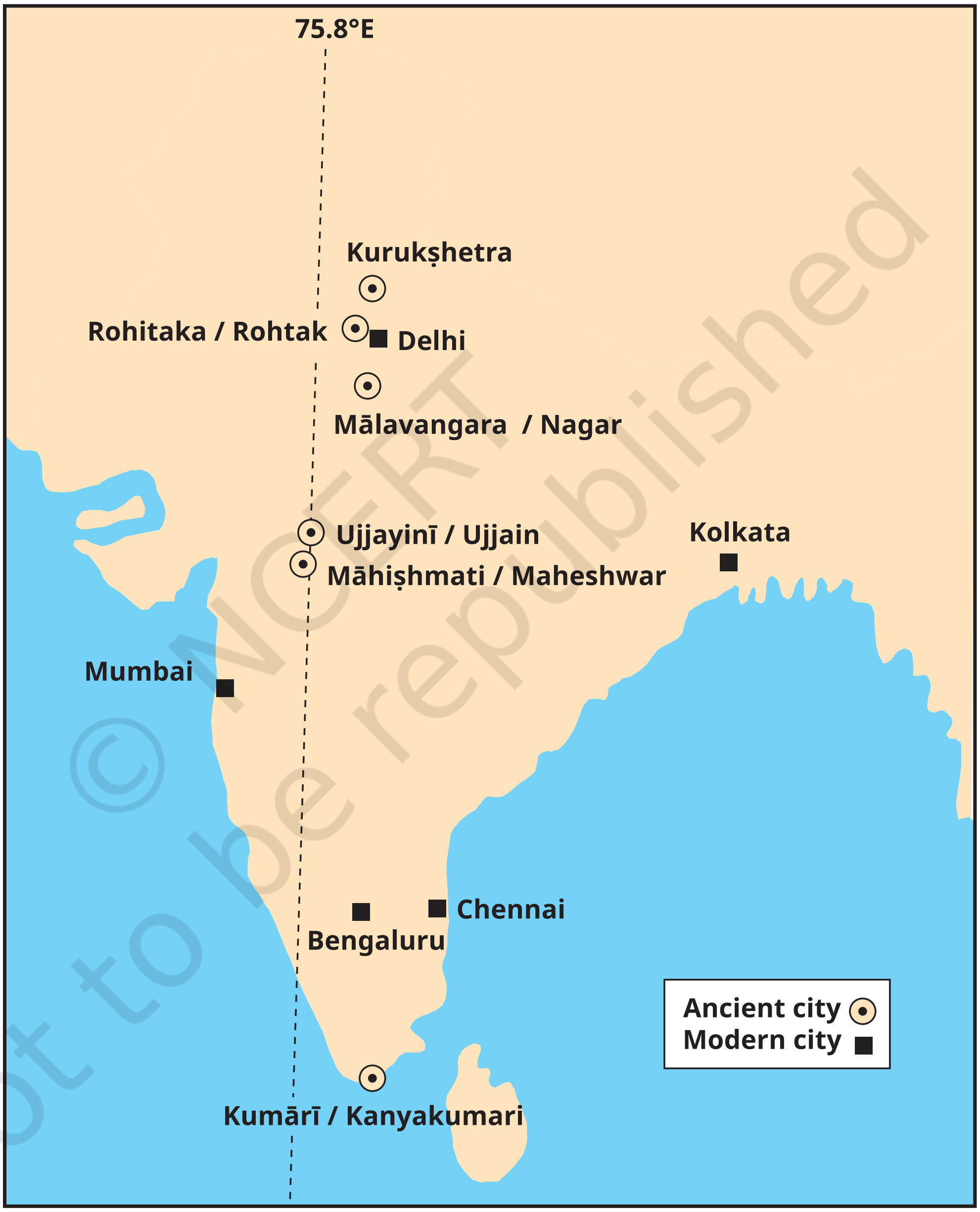


Fig. 1.5. The Ujjayinī prime meridian used in ancient Indian astronomy. Cities marked with a circle are mentioned in astronomical texts as being on this meridian (the modern name is given after the oblique bar).

Reprint 2025-26

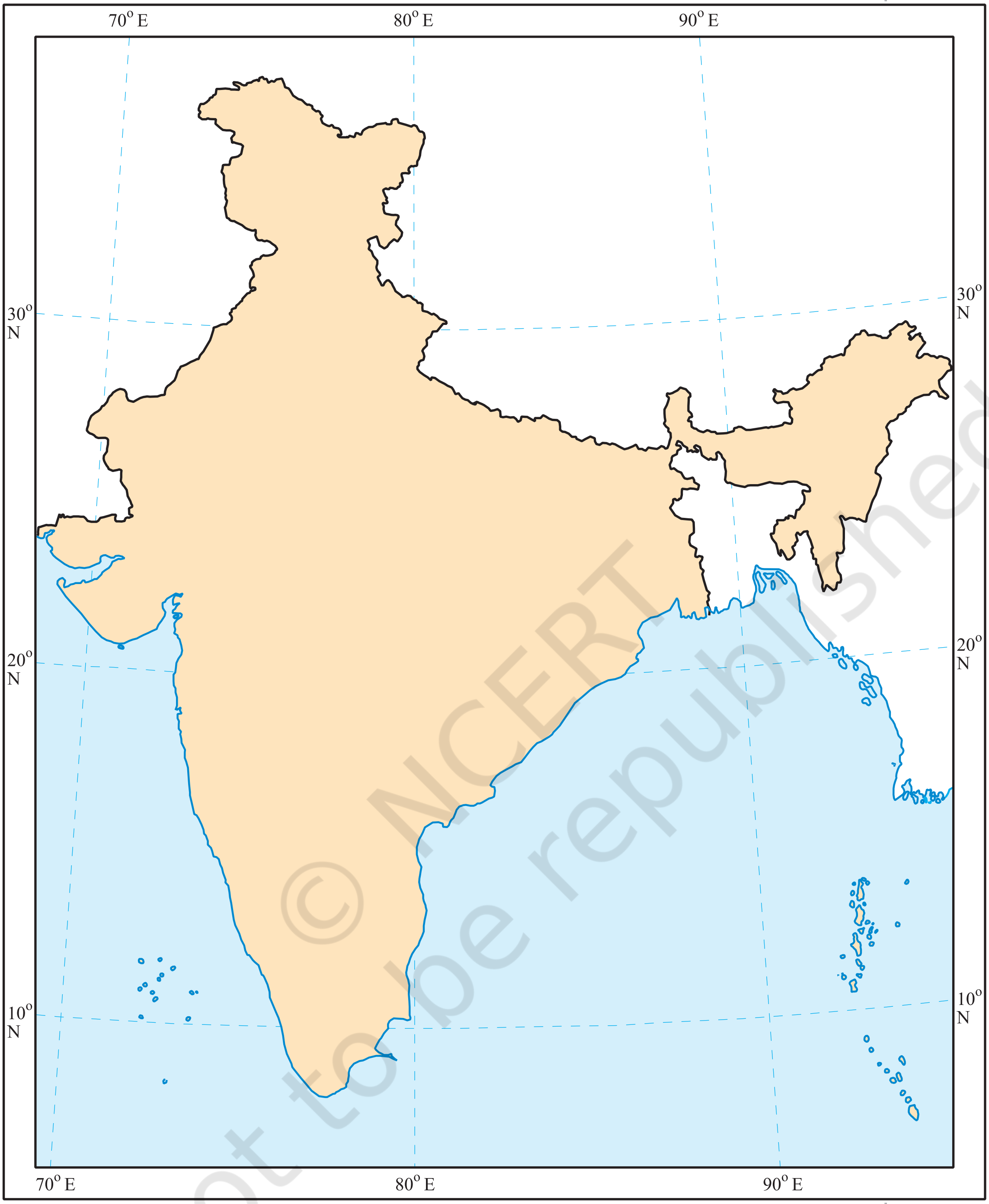


Fig. 1.6. This map, published by the Survey of India, shows the country's outline along with a few parallels of latitude and meridians of longitude. India's latitudes extend approximately from to 8°N to 37°N, and longitudes approximately from 68°E to 97°E. (The two colours have been added.)

Understanding Time Zones

Let's make the globe rotate again from west to east — that is how our planet spins around its axis, making a full turn every 24 hours. A full turn is 360°, so this means 15° per hour (15 × 24 = 360). Let us now mark the meridians of longitude every 15°. Moving eastward from the Prime Meridian, we get 0°, 15°E, 30°E, 45°E, and so on every 15° up to 180°E. It is the same as adding one hour of **local time** with each meridian — if it is 12 pm or noon at Greenwich, it is 1 pm local time at 15°E, 2 pm at 30°E, and so on. But going westward, it is the other way round — 11 am local time at 15°W, 10 am at 30°W, etc.

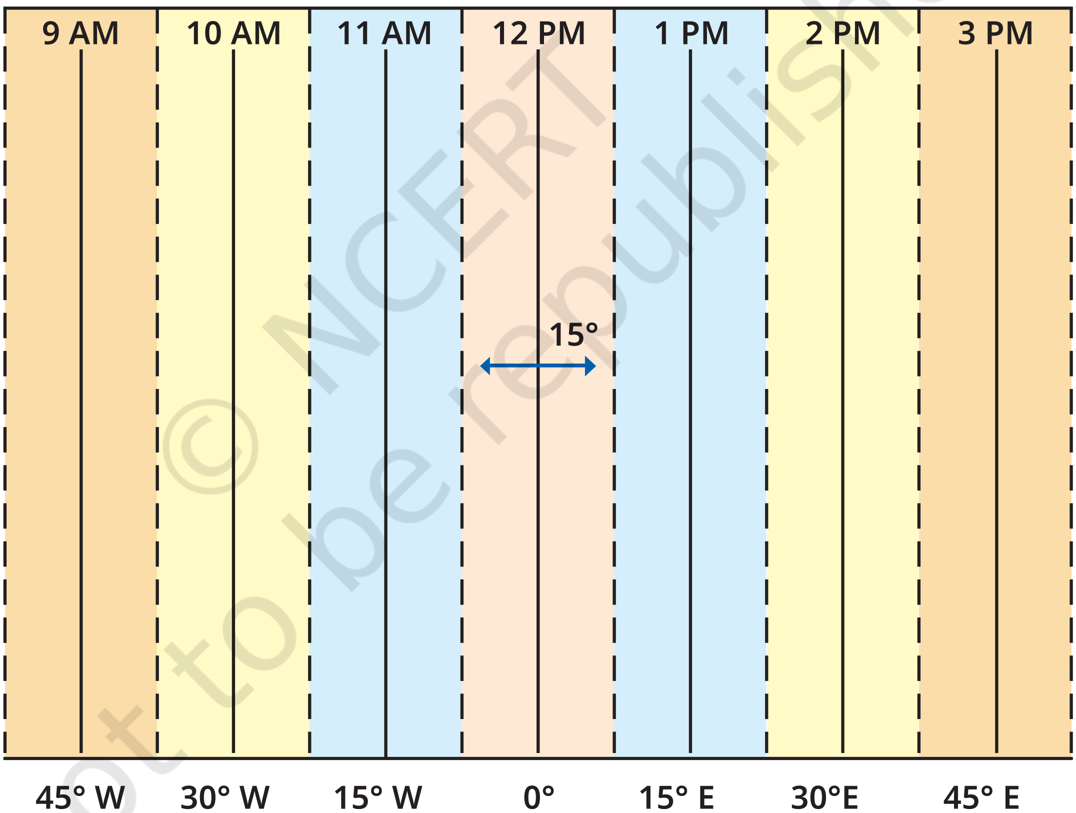


Fig. 1.7. This graph shows longitudes at the bottom and the local time at the top, with reference to the Prime Meridian at 0°. Each colour is a zone of 15° centred on a meridian.

Reprint 2025-26

LET'S EXPLORE

Two friends, one sitting in Porbandar (Gujarat) and the other in Tinsukia (Assam), are speaking on the phone late afternoon. The latter remarks that the sun has set in Assam and it's now dark. The former is surprised and says, "But it's still full daylight here!" Explain why. And, as a class activity, calculate the difference in local time between those two cities. *(Hint: for now, consider the difference in longitude between Porbandar and Tinsukia to be 30°; later, you can find out the precise value.)*

The same method can be used to calculate the local time of any place on the Earth. But it would not be convenient for a country to use many local times! That is why most countries adopt a **standard time** based on a meridian passing through them. Indian Standard Time (IST) is 5 hours 30 minutes (also noted 5.5 hours) ahead of the local time at Greenwich (called **Greenwich Mean Time** or GMT).

LET'S EXPLORE

Return to the two friends sitting in Gujarat and Assam. Use this example to explain the difference between local time and standard time.

All these standard times are organised in time zones, which broadly follow the zones of 15° in the graph (Fig. 1.7). But let us consider the world map below (Fig. 1.8). We can see that the lines dividing the time zones are not fully straight. This is because they have to respect each country's standard time and, therefore, tend to follow international borders. The numbers written inside some countries are the numbers of hours to be added to GMT to get their standard times if they have a positive sign, or subtracted from GMT if they have a negative sign.

Reprint 2025-26

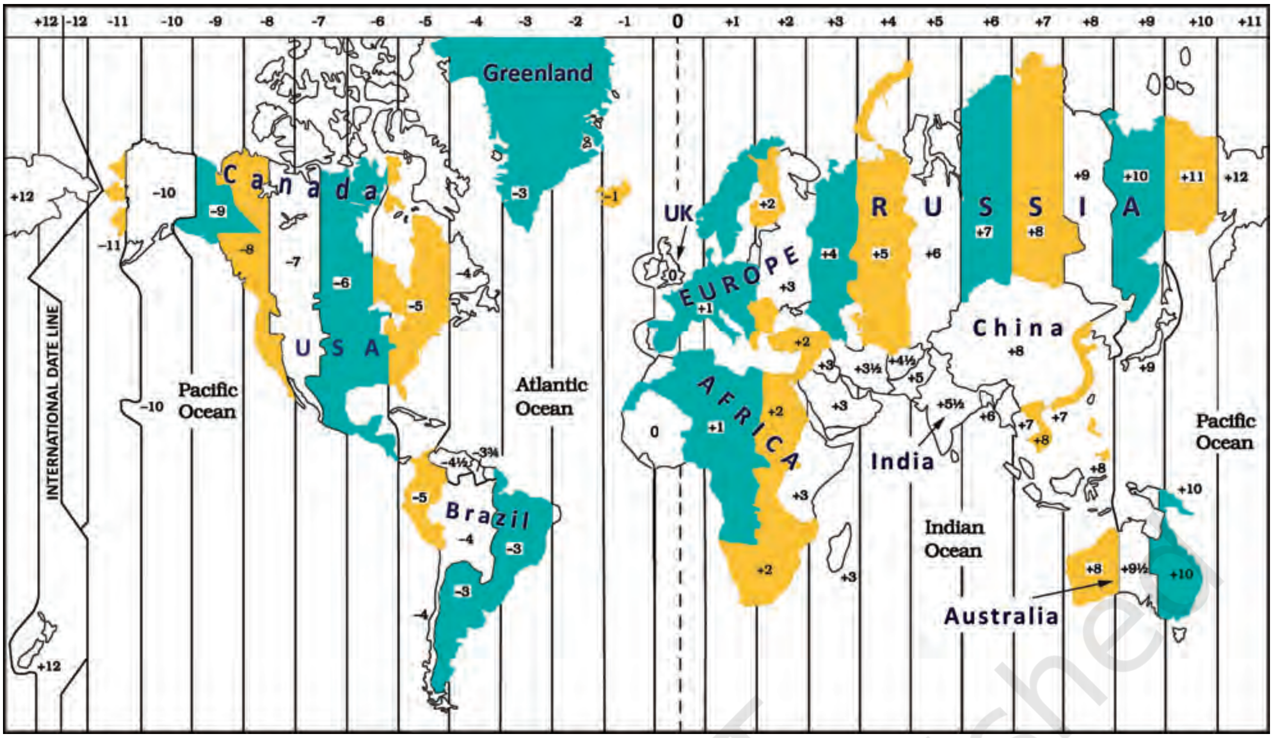


Fig. 1.8. A world map of the time zones, showing the standard times (with respect to GMT) for a few countries. (Note that international borders are approximate, not exact.)

DON'T MISS OUT

From the above explanation, it may seem as if every country has one standard time. That is not always the case. Some countries, like Russia, Canada or the USA, are too large to have a single time zone. The USA has six time zones and Russia has 11 — which means that travelling across Russia from east to west, you will need to readjust your watch 10 times to align with the local time!

Similarly, the globe in Fig. 1.9, centred on India, shows standard times with respect to GMT for a few countries.

Finally, while the Prime Meridian was fixed at Greenwich, the opposite line — at approximately 180° longitude — is called the **International Date Line**.

As you can see on the map, the +12 and the –12 time zones touch each other at this line. If you cross it by ship or plane, you need to change the date in your watch. If you cross it

Reprint 2025-26

Fig. 1.9 A few time zones (with respect to GMT) in Africa and Eurasia.

travelling eastward, you subtract a day (say, from Monday to Sunday); if you cross it travelling westward, you add a day (from Sunday to Monday). We said that the International Date Line is 'approximately' at 180° longitude, as it deviates in places to avoid dividing some countries into two different days!

Before we move on ...

- Maps are a very useful tool to represent an area of the Earth, whether small or large. The main components of maps are distance, direction and symbols.
- Every place on the Earth has a location which can be precisely defined with the help of a grid of latitudes and longitudes — imaginary lines running from east to west (parallel to the Equator) and north to south (from pole to pole) respectively.

Reprint 2025-26

- Longitude also marks the time and defines the time zones.
- The International Date Line is located approximately at 180 degrees longitude, opposite the Prime Meridian. Crossing the International Date Line changes the date by one day.

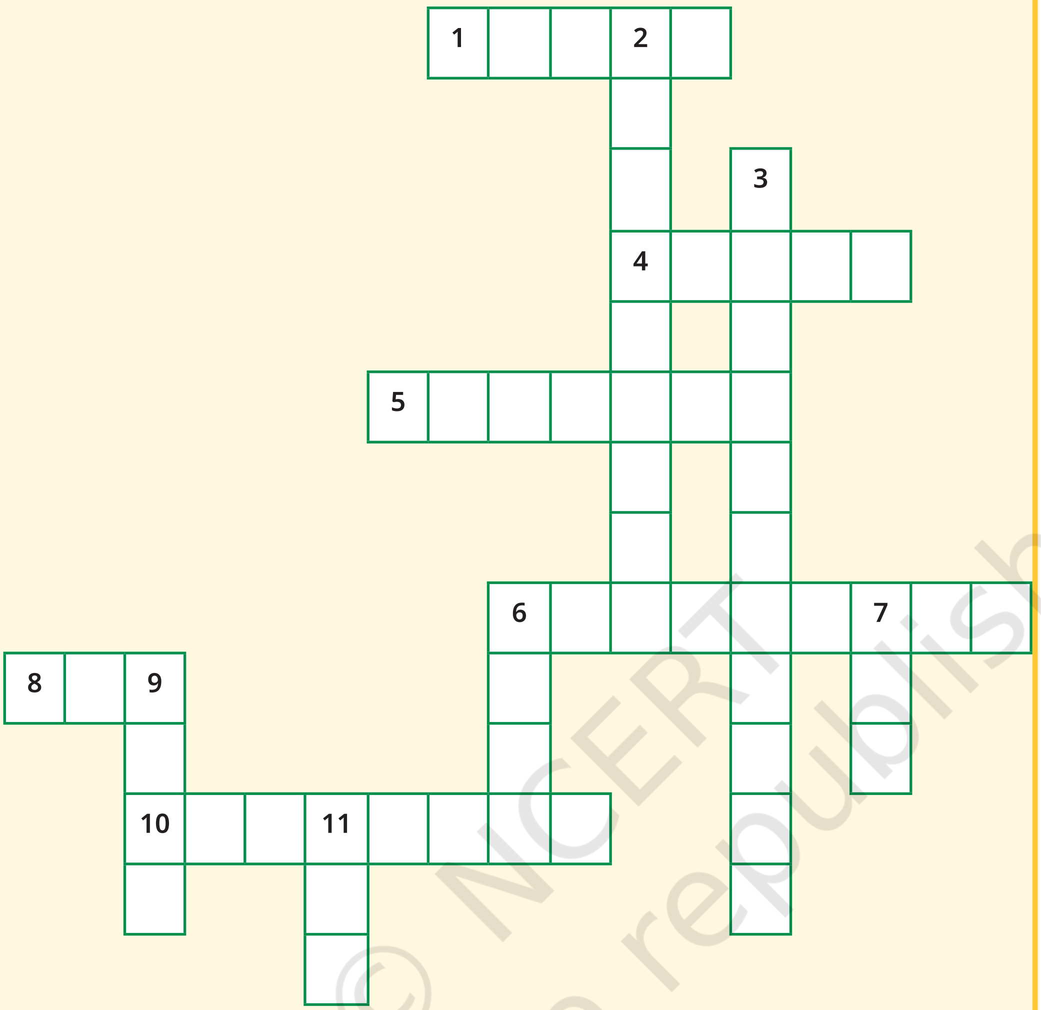
Questions, activities and projects

Estuary: The place where a river meets the sea.

1. Returning to page 10 and to Fig. 5.2 in Chapter 5 of this textbook, taking the scale to be 2.5 cm = 500 km, calculate the real distance from the **estuary** of the Narmada River to the estuary of the Ganga river. ***(Hint: round off your measurement on the map to an easy number.)***
2. Why is it 5:30 pm in India when it is 12 pm or noon in London?
3. Why do we need symbols and colours in the map?
4. Find out what you have in the eight directions from your home or school.
5. What is the difference between local time and standard time? Discuss it in groups, with each group writing an answer in 100 to 150 words. Compare the answers.
6. Delhi's and Bengaluru's latitudes are 29°N and 13°N; their longitudes are almost the same, 77°E. How much will be the difference in local time between the two cities?
7. Mark the following statements as true or false; explain your answers with a sentence or two.
   - All parallels of latitude have the same length.
   - The length of a meridian of longitude is half of that of the Equator.
   - The South Pole has a latitude of 90°S.
   - In Assam, the local time and the IST are identical.
   - Lines separating the time zones are identical with meridians of longitude.
   - The Equator is also a parallel of latitude.
8. Solve the crossword below.

Reprint 2025-26

Locating places on Earth



Across

1. Lets you squeeze a huge area into your map
4. A convenient sphere
5. The longest parallel of latitude
6. The place the Prime Meridian is attached to
8. So convenient to find your way
10. A measure of the distance from the Equator

Down

2. A measure of the distance from the Prime Meridian
3. These two together allow us to locate a place
6. What latitudes and longitudes together create
7. The time we all follow in India
9. On top of the world
11. An abbreviation for a line across which the day and date change

Reprint 2025-26

Noodles

*'Noodles' is our abbreviation for 'Notes and Doodles'!

Reprint 2025-26

CHAPTER 2

Oceans and Continents

> *The ocean is everything. It covers seven-tenths of the terrestrial globe. Its breath is pure and healthy. It is an immense desert, where man is never lonely, for he feels life stirring all around. ... The ocean is the vast reservoir of Nature. The globe began with the ocean, so to speak, and who knows if it will not end with it. ...*
>
> — *Jules Verne (1870)*

Fig. 2.1 The Earth seen from space (photograph by the Lunar Reconnaissance Orbiter). The view is centred on the Pacific Ocean, with Africa to the left, India and part of Asia at the top, Australia to the right, and Antarctica at the bottom.

The Big Questions?

1. *What are oceans and continents? What are their names and their distribution?*
2. *In what ways do oceans and continents impact life on Earth, including human life?*

0681CH02

Let us return to our globe and rotate it gently. Or look at the picture of the Earth seen from the Moon. What is the most widespread colour you see? Blue, obviously, but what does it represent? You must have guessed the answer — it is ‘water’. This means that most of the Earth’s surface is actually covered with water — almost three-fourths of the surface, in fact. That is why, when seen from outer space, the Earth appears mostly blue. Indeed, early astronauts lovingly called the Earth the ‘blue planet’.

The largest water bodies we see on the globe are called ‘**oceans**’.

But in the picture of the Earth (Fig. 2.1), you can see at least one other colour, brown. This colour is that of land, which covers a little over one-fourth of the globe. A large body of land is called a ‘**landmass**’, and a large continuous expanse of land is called a ‘**continent**’.

Both oceans and continents play a vital role in shaping the climate of the Earth. They affect all aspects of life, including all plants and animals, and therefore, human life too. We see their impact throughout our history and culture, and in our daily lives.

DON’T MISS OUT

The emblem of the Indian Navy contains the motto *Sam noh Varunah* (pronounced ‘*Śham no Varuṇah*’), which means, “Be auspicious to us, O Varuna.” This is an invocation to Varuṇa, a Vedic deity associated with the oceans, the sky, and water in general.

Reprint 2025-26

The Distribution of Water and Land on the Earth

As it happens, oceans and continents are not distributed equally between the Northern and Southern Hemispheres.

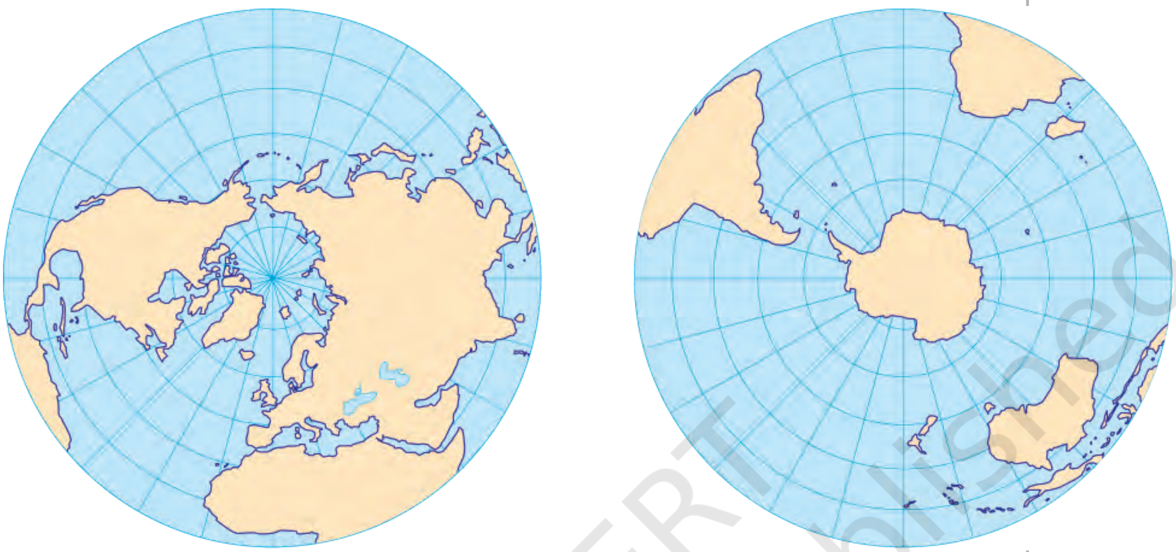


Fig. 2.2. Maps of the Earth as seen from above the North Pole (left) and above the South Pole (right).

Let us examine the two maps in Fig. 2.2. Here too, the blue areas consist of oceans, along with their smaller extensions, which have various names — '**sea**', '**bay**', '**gulf**', etc.

Definitions for these terms are in the Glossary at the end of this textbook.

LET'S EXPLORE

→ What are the circular lines in each map called? And do you know what the lines radiating out of the two poles are called? (*Hint: you studied them in the previous chapter, but here they are presented differently.*)

→ Which hemisphere holds more water?

→ What do you think could be the approximate proportion of water to land in the Northern Hemisphere? And in the Southern Hemisphere? Discuss in groups.

→ Are all the oceans connected with one another, or are there separations between them?

Reprint 2025-26

Ocean Life

Coral reef

A star fish on a sea anemone

Shark

Sperm whale, mother and baby

Dolphins

Shallow coral reef with colourful tropical fish

Algae

Emperor penguins

Bonaire sea turtle

The oceans together hold most of the water available on the planet. But this seawater is salty and unfit for consumption by most land animals, including humans. On the other hand, freshwater makes up a very small proportion of the planet's water resources; it is found in glaciers, rivers, lakes, in the atmosphere and also underground (the last is called 'groundwater').

THINK ABOUT IT

- If there is such abundance of water on the planet, why is there so much talk of 'water scarcity' or a 'water crisis'?
- What ways of saving water are you aware of? Which ones have you seen practised at home, at your school, and in your village, town or city?

Oceans

On the world map in Fig. 2.3 on page 32, we can observe five oceans — the Pacific Ocean, the Atlantic Ocean, the Indian Ocean, the Arctic Ocean and the Southern (or Antarctic) Ocean.

Although we have listed five oceans, it is clear from the map that they are not really separate. The lines that divide them on the map are no more than conventions — the natural world does not follow such boundaries. Seawater, for example, constantly flows across different oceans, sustaining a rich diversity of **marine** life. Many plant and animal species can be found across multiple oceans.

The marine **flora** includes tiny plants called algae and all kinds of seaweeds; the marine **fauna** consists of thousands of species of colourful fish, dolphins, whales, and countless mysterious deep-sea creatures. Each part of the ocean, from the sun-lit surface to the dark depths, has its own diverse life forms.

Marine: Related to or found in the oceans and seas.

Flora: The plant life of a particular region or period of time.

Fauna: The animal life of a particular region or period of time.

Reprint 2025-26

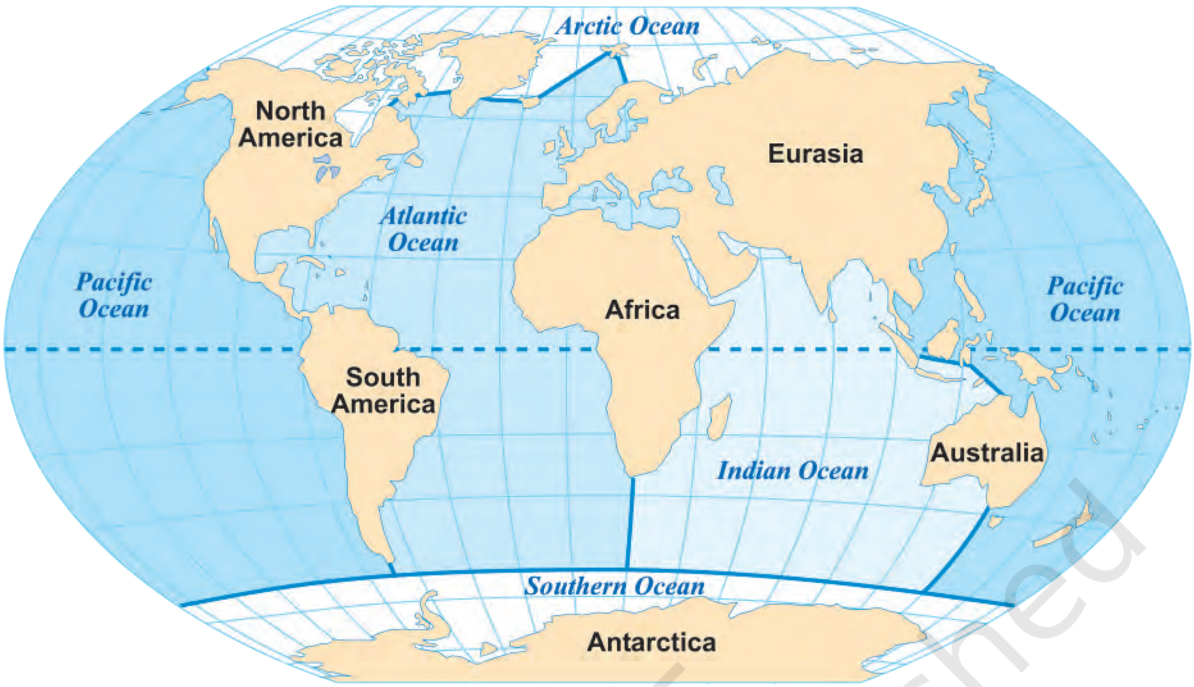


Fig. 2.3. A world map showing the five oceans, their conventional boundaries, and the continents

LET'S EXPLORE

Locate the five oceans and, in the table below, mark the hemisphere or hemispheres they belong to.

	Northern Hemisphere	Southern Hemisphere
Pacific Ocean		
Atlantic Ocean		
Indian Ocean		
Southern Ocean		
Arctic Ocean		

It is visible on the map that the Pacific Ocean is the largest of all, followed by the Atlantic Ocean. The Indian Ocean is the third largest, while the Southern Ocean is the fourth. The smallest one is the Arctic Ocean.

Reprint 2025-26

DON'T MISS OUT

- As the map of oceans makes clear, the main limits of the Indian Ocean are Asia to the north, Africa to the west and Australia to the east, apart from the Southern Ocean in the south.
- On either side of India, we find two parts of the Indian Ocean — the Arabian Sea to the west and the Bay of Bengal to the east.

Fig. 2.4 (on the right). This map of India is the same as Fig. 1.6, but with the addition of the Arabian Sea and the Bay of Bengal. Also marked are India's two major groups of islands (see subsection on 'Islands' further below).

Oceans and disasters

Returning to the picture of the Earth at the start of this chapter, you may have noticed white shapes across the globe. Did you guess what they are? They are large masses of clouds. Such clouds bring rain to the continents; for instance, the monsoon rains we in India expect every summer originate in the ocean — without such rains, our agriculture and all life will suffer. But oceans often also give rise to storms — violent events with extreme rainfall or very strong winds, such as cyclones, which can cause widespread damage to coastal regions of the world.

A tsunami is another natural disaster that originates in the ocean. It is a huge and powerful wave generally caused by a strong earthquake or a volcanic eruption at the bottom of the ocean. Tsunamis can travel thousands of kilometres and submerge coastal areas, causing widespread damage.

DON'T MISS OUT

- On 26 December 2004, India and another 13 countries around the Indian Ocean were struck by a powerful tsunami caused by an earthquake in Indonesia. More than two lakh people lost their lives. In India, the Andaman and Nicobar Islands (see Fig. 2.4 above, and also the subsection 'Islands' further below) and the coasts of Tamil Nadu and Kerala were severely affected and suffered much damage and loss of life.
- Such tsunamis are rare but very destructive. Luckily, they can often be detected before they hit a coast. Many countries collaborate in such 'early warning systems'. There is, in particular, an Indian Ocean Tsunami Warning System, to which many countries, including India, contribute. This helps to take measures to protect lives and property.
- Events that lead to loss of life and property are handled under **disaster management**. India has its own 'National Disaster Management Authority' to deal with all kinds of disasters (we will see more examples in the next chapter).

Continents

Continents are visible on the map of oceans (Fig. 2.3). How many can you count? The answer is not so simple, as they can be counted in several ways. Depending on our choice, we may list any number of continents between four and seven! Here is why:

- North America and South America are generally considered to be two continents; but if seen as a single landmass, they can also be considered as one.
- Europe and Asia are generally considered as two continents, although the map makes it clear that they form a single landmass. For historical and cultural reasons, Europe's evolution has been very different

Reprint 2025-26

from Asia's, which is why they can be seen as two continents. Geologists, however, often regard them as a single continent called 'Eurasia'.

- Africa and Eurasia are generally regarded as two continents, but sometimes as one.

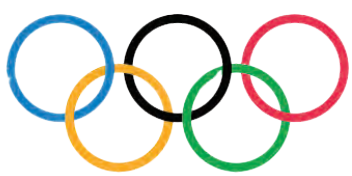
Let us summarise the different counts in a table:

Count of continents (in alphabetical order)	
Four continents	Africa-Eurasia, America, Antarctica, Australia
Five continents	Africa, America, Antarctica, Australia, Eurasia
Six continents	Africa, Antarctica, Australia, Eurasia, North America, South America *(this is reflected in Fig. 2.3 on page 32)*
Seven continents	Africa, Antarctica, Asia, Australia, Europe, North America, South America

In practice, the last list of seven continents is the one most widely adopted and used.

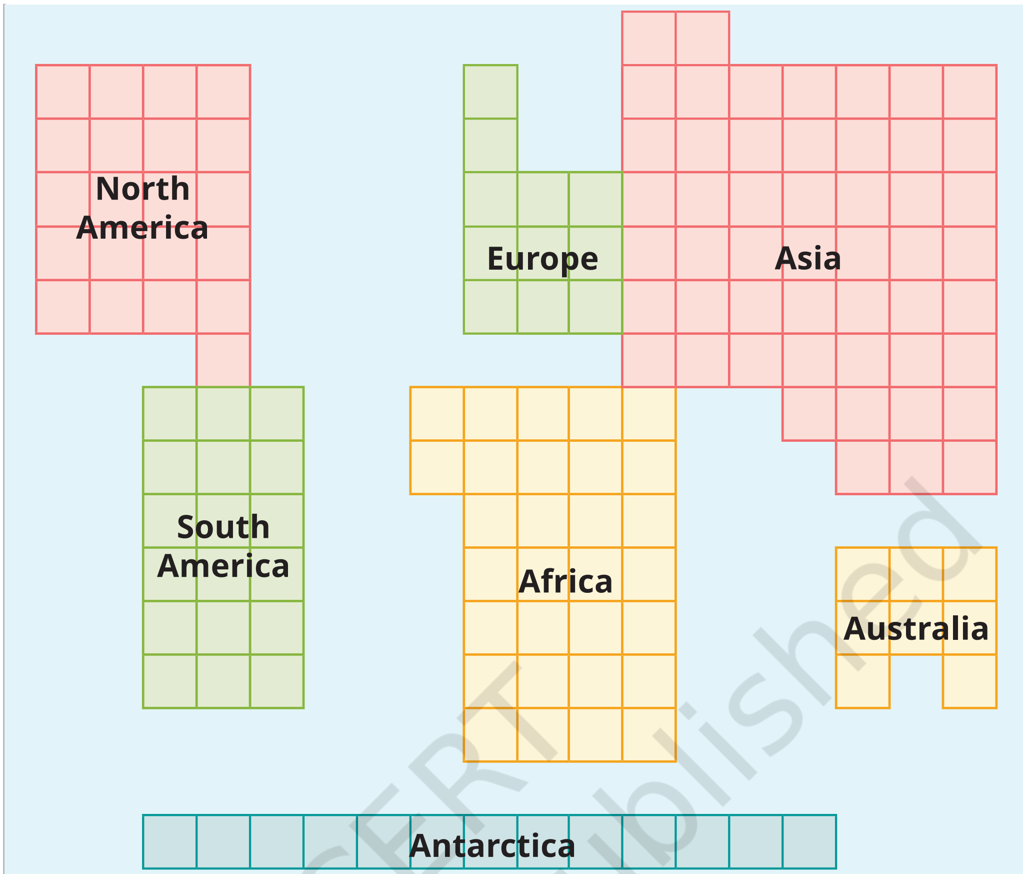
DON'T MISS OUT

You may have seen the five Olympic rings, one of the symbols of the Olympic Games. They symbolise the gathering of sportspeople from all over the world. The rings were chosen to represent five inhabited continents — Africa, America, Asia, Australia and Europe.



Now let us look at the diagram on page 36, which is based on the list of seven continents. It does not show their actual shapes, but their relative sizes.

Reprint 2025-26



LET'S EXPLORE

- Counting the numbers of squares, name the largest continent and the smallest.
- Which one is larger — North America or South America? Africa or North America? Antarctica or Australia?
- Re-colour the diagram by having a single colour for Europe and Asia and rename the result as 'Eurasia'. Compare its size with South America's.
- Write down the list of continents from the smallest to the largest.

Islands

If you have carefully observed the two maps earlier in this chapter (Fig. 2.2 and 2.3), you may have noticed that

Reprint 2025-26

the continents do not include all landmass. Some smaller pieces of land are left out; surrounded by water on all sides, they are called **islands**. (Continents are also surrounded by water, but because they are so large, they are not considered islands.)

There are lakhs of islands on the planet, of very different sizes.

DON'T MISS OUT

- Greenland is the largest island in the world (locate it on a globe or a map). You would have to add the areas of the 10 largest states of India to reach its size.
- India has more than 1,300 small islands! Those include two major groups — Andaman and Nicobar Islands in the Bay of Bengal and Lakshadweep Islands in the Arabian Sea (see Fig. 2.4).
- Since 1981, the Indian Antarctica Programme has been exploring Antarctica, a continent with a very cold climate and harsh environment (see the white expanse at the bottom of Fig. 2.1, which is mostly ice). In 1983, India established its first scientific base station there, called 'Dakshin Gangotri' (two more bases were established later). About 40 teams of Indian scientists have conducted research in this faraway region, especially on the evolution of climate and environment. The settlement where the scientists live has a library and even a post office!

Oceans and Life

Oceans and continents are vital parts of the environment and affect most aspects of our lives, even if we do not notice it. We have mentioned that oceans send rain to the continents; this is part of the Earth's water cycle, which you will further study in Science. Without oceans, for

Reprint 2025-26

instance, there would be no rainfall! The Earth would be a desert. Moreover, more than half of the world's oxygen is produced by the oceans' flora, which is why they are called 'the planet's lungs'. The oceans, therefore, play a crucial role in regulating the climate and sustaining life on Earth.

Oceans have deeply impacted humanity in many other ways. From early times, people have used oceans and seas to migrate to other regions, to trade in all kinds of goods, to conduct military campaigns, and as a source of food through fishing. Oceans have also nourished the cultures of coastal people all over the world. Almost all of them have tales and legends about the sea, sea gods and goddesses, sea monsters and treasures from the sea — the oceans' dangers but also their blessings.

DON'T MISS OUT

The United Nations has designated June 8 as World Oceans Day to "remind us all of the major role the ocean plays in everyday life. It serves as the lungs of our planet, a major source of food and medicine and a critical part of the biosphere." Scientific studies have shown how the oceans are polluted by human activity — we throw several million tonnes of plastic waste into the oceans every year, choking marine life. There are several other forms of pollution. As a result, the marine environment is under threat. Overfishing (excessive fishing) is another cause for the decline of marine life. It is our collective responsibility to protect oceans for the future of the planet and of humanity.

Before we move on ...

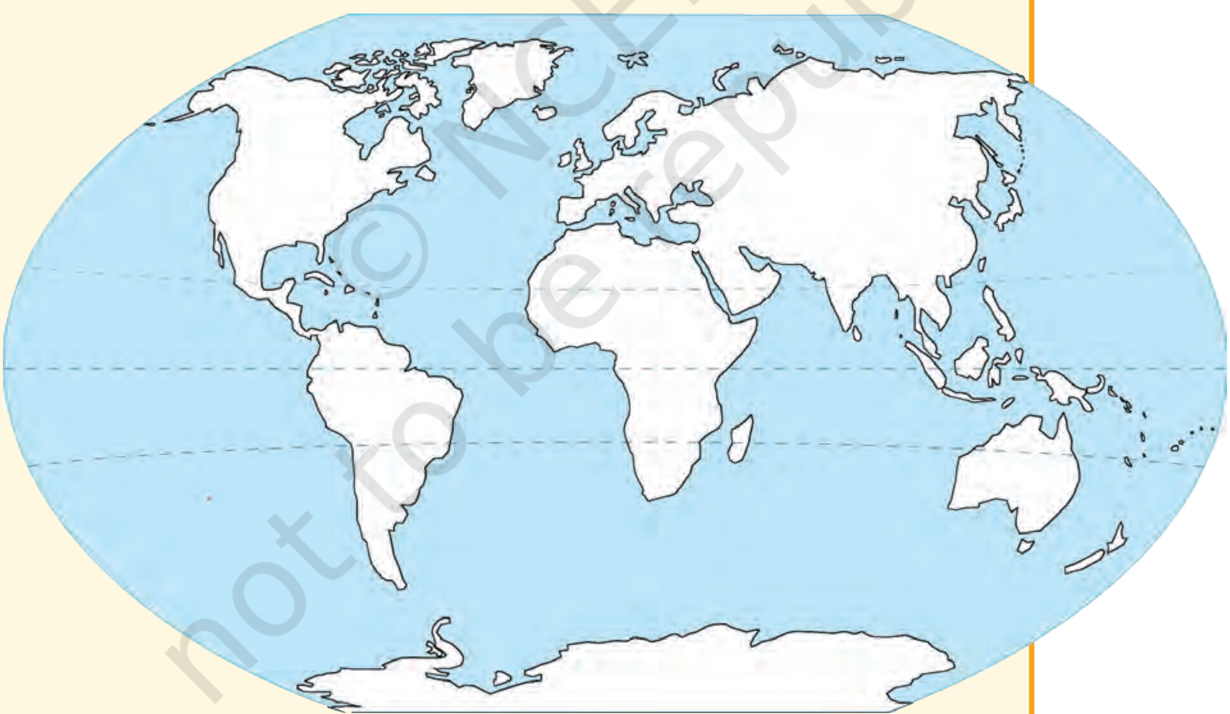
- The Earth's surface has vast water bodies called 'oceans' and large landmasses called 'continents'. Oceans are interconnected. Continents may be counted in various ways; the most common count is seven.
- The Northern Hemisphere has more land than the Southern Hemisphere.

Reprint 2025-26

- Oceans support all kinds of marine life and play a critical role in the world climate. They are now seriously affected by human activity and need our collective protection.

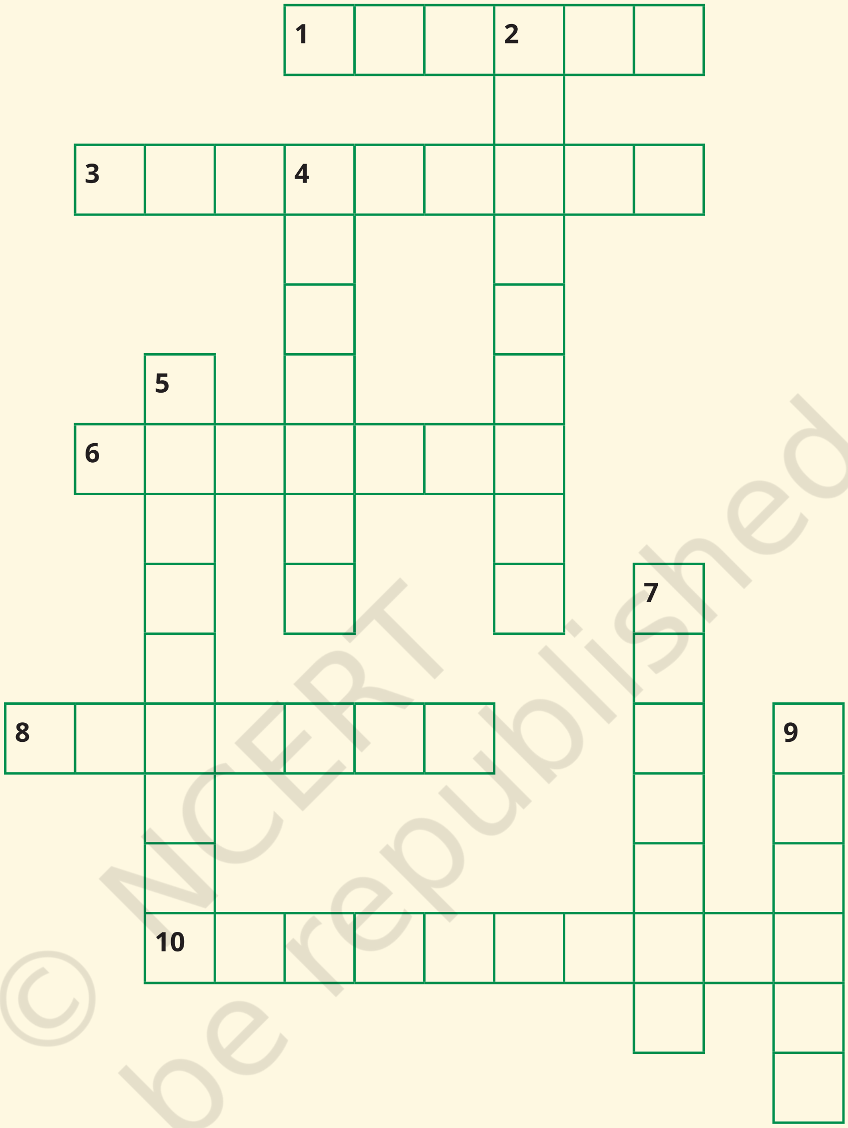
Questions, activities and projects

1. Explain the following terms:
   (a) Continent
   (b) Ocean
   (c) Island
2. Let us draw – Without looking at the maps in this chapter, draw the continents free hand on a sheet of paper and colour them. Then compare your drawing with the map of oceans and continents in the chapter.
3. Let us do – On the outline map of the world given below, label all the continents and oceans.



Reprint 2025-26

4. Solve this crossword



Across

1. Abundantly produced by the oceans
3. A large expanse of landmass
6. A large continent of which India is a part
8. A major source of pollution of the oceans
10. The coldest continent

Down

2. The largest island on Earth
4. A huge destructive wave from the ocean
5. The smallest continent
7. The largest body of water on the Earth
9. A landmass (but not a continent) surrounded by the sea or ocean

Reprint 2025-26

CHAPTER 3

Landforms and Life

Free from the burden of human beings, may the Earth with many heights, slopes and great plains, bearing plants endowed with varied powers, spread out for us and show us her riches! … The Earth is my mother and I am her child.

— Atharva Veda, Bhūmi Sūkta ('Hymn to the Earth')

The Big Questions ?

1. *What are the major types of landforms and their significance to life and culture?*
2. *What are the challenges and opportunities of life associated with each landform?*

0681CH03

Introduction

Humans, like most mammals, live on land. Land, as you may have noticed, has many forms and features; its appearance changes a lot from one region to another. Suppose that you are travelling by road from the region known as Chhota Nagpur in Jharkhand, reach Prayagraj in Uttar Pradesh, and go on to Almora in Uttarakhand. On the way, you will see very different landscapes. In fact, you will encounter three major landforms, which we will now explore.

LET'S EXPLORE

- → As a class activity, form groups of four or five students and observe the school's surroundings. What kind of landscape do you see? Will the landscape change a few kilometres away? Or within some 50 kilometres? Compare with other groups.
- → In the same groups, discuss a journey that any of you has made through a region of India. List the different landscapes seen on the way. Compare with other groups.

A landform is a physical feature on the surface of our planet Earth. Landforms take shape over millions of years and have a significant connection with the environment and life. They can broadly be divided into three categories — **mountains, plateaus and plains** (Fig. 3.1).

These landforms have different climates and are home to a variety of flora and fauna. Humans have adapted to all landforms, but the number of people living on different kinds of landforms varies throughout the world.

Altitude: The height of an object above sea level. Examples: the altitude of a mountain, the altitude of a bird or plane in flight, the altitude of a satellite.

Mountains

Mountains are landforms that are much higher than the surrounding landscape. They can be recognised by a broad base, steep slopes and a narrow summit. Depending on their height, some mountains are covered with snow. At lower **altitudes**, the snow melts every summer and turns

Reprint 2025-26

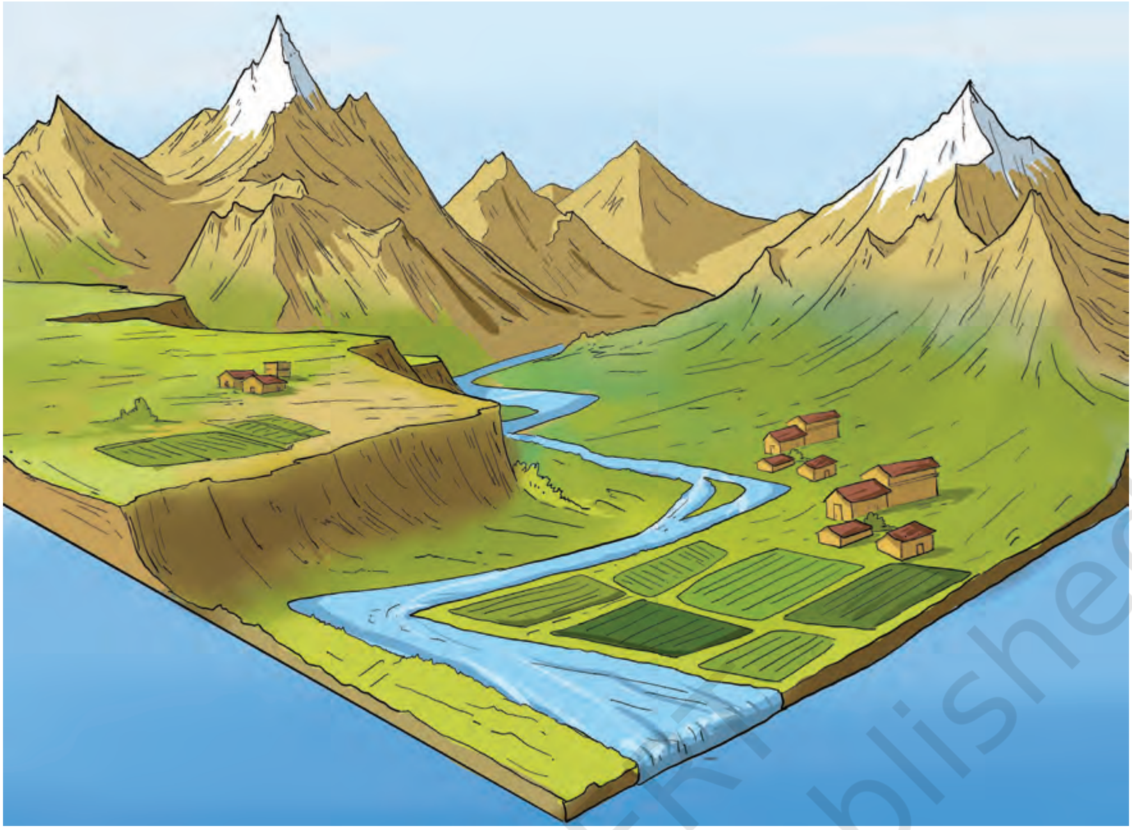


Fig. 3.1. This drawing illustrates three landforms — mountains in the background (two of them snow-capped), a plateau on the left and a plain in the foreground, with a river emerging from the mountains.

into water that feeds rivers. At high altitudes, the snow may never melt, leaving the mountain permanently snow-capped.

Other highlands with a lower height, less steep slopes and rounded tops are called **hills**.

Precipitation: Water from the atmosphere reaching the ground in any form — rain, snow and hail are the most common forms of precipitation.

THINK ABOUT IT

What is snow? Unless you live in a Himalayan region (such as Kashmir, Ladakh, Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh), you may never have seen snow! In the rest of India, most **precipitation** is in the form of rain and hail. But at higher altitudes, if it is cold enough, snow will fall, covering the landscape in a soft and beautiful white blanket. Snow and hailstones are nothing but precipitation of water in a solid state.

Reprint 2025-26

Fig. 3.2. Pictures of six mountains of the world

Most of the world's mountains are grouped in **mountain ranges,** such as the Himalayas in Asia, the Alps in Europe and the Andes in South America. Some of these ranges stretch for thousands of kilometres.

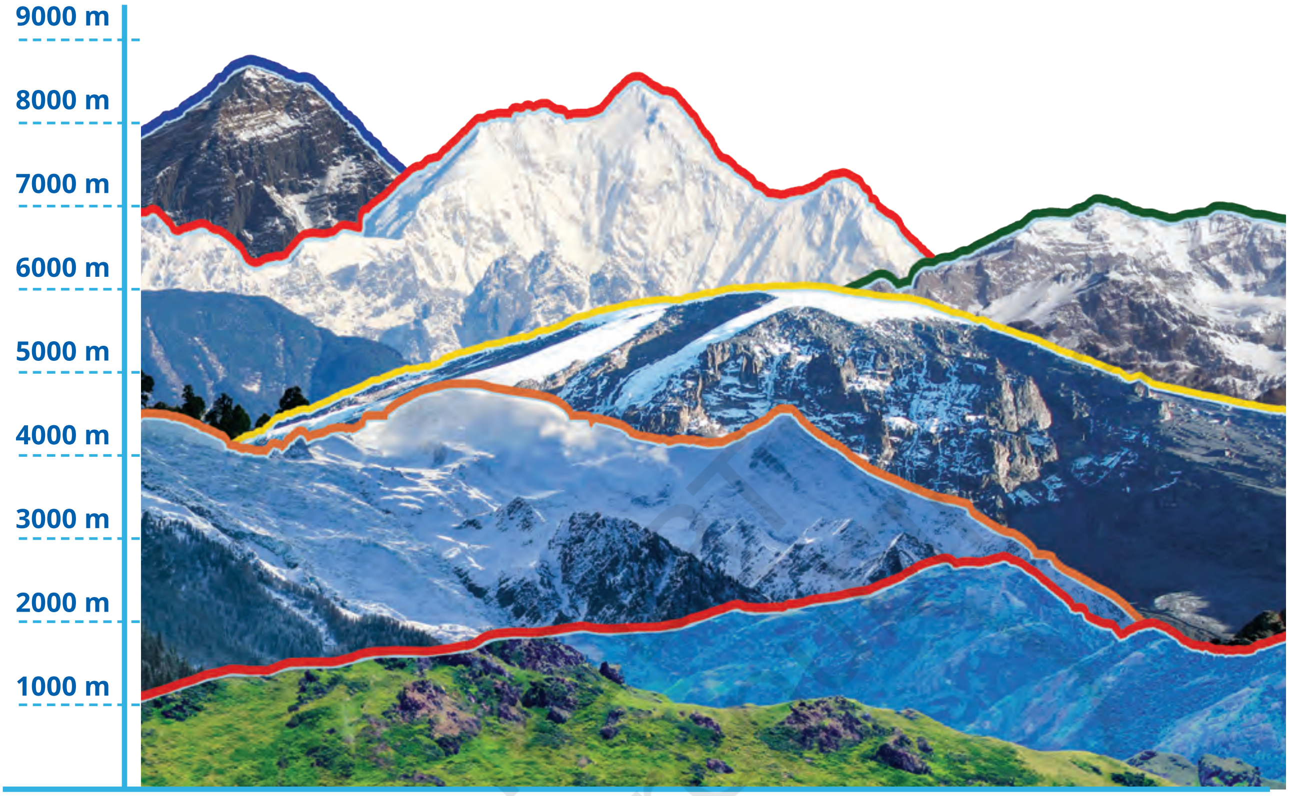


Fig. 3.3. A sketch showing the relative heights of six mountains of the world

Fig. 3.2 shows pictures of six mountains of the world. Fig. 3.3 brings them together to give a visual impression of their relative heights from top to bottom. Mount Everest (between Tibet (China) and Nepal) and Kanchenjunga (between Nepal and the Indian state of Sikkim) are the two highest peaks of the Himalayan range. Mount Aconcagua (in South America) is the highest peak of the Andes. Mount Kilimanjaro in eastern Africa is an isolated mountain that is not part of any range. Mont Blanc in Western Europe is the highest mountain of the Alps. Anamudi (in Kerala, also known as 'Anai Peak') is the highest mountain in south India.

Reprint 2025-26

Montane forest: A type of forests that grows in mountainous regions.

Moss: A small green plant without flowers or true roots, often spreading in a cushion-like cover.

Lichen: A plant-like organism that generally clings to rocks, walls or tree.

Mountains with tall and sharp peaks, like the Himalayas, are relatively 'young', which means that they were formed recently in the Earth's history — but that is still millions of years ago! Shorter and more rounded mountains and hills, like the Aravalli Range, are much older and have been rounded by erosion. Sometimes, as with the Himalayas, upliftment as well as erosion continue to this day. (You will learn more in Science about such processes and their causes; let us just say here that some mountains of the world, like the Himalayas, are still growing in height.)

Mountain environment

Mountain slopes are often covered with a type of forest called **montane forest,** where conifer trees such as pines, firs, spruce and deodar are common. These conifer trees grow tall and cone-shaped, with thin, pointed leaves. At higher altitudes, the trees give way to grasses, **mosses** and **lichen.**

Here are two verses from a long poem by Kālidāsa, who lived at least 1,500 years ago and is often considered to be the greatest poet of ancient India. The poem, *Kumārasambhava*, begins with an invocation to the Himalayas. (This is a simplified translation from the Sanskrit.)

> *In the north rises Himālaya, the Lord of mountains, like a living god, who measures the Earth and stretches from the western to the eastern oceans. …*
>
> *From it the wind comes down, carrying spray from descending Gangā, shaking the deodar trees, opening the peacocks' tail feathers and cooling the mountain people after they hunt deer.*

Discuss the verses and the following questions in class.

- What are the 'western to the eastern oceans'? Can you locate them as well as the 'Lord of mountains' on Fig. 5.2?
- Why is Gangā mentioned? *(Hint: There could be several reasons.)*

Reprint 2025-26

Fig. 3.4. A few mountain animals

Deep forests, flowing rivers, lakes, grasslands and caves in the mountains are home to diverse fauna, for instance, the golden eagle, the peregrine falcon, the Canadian lynx, the snow leopard, the ibex, the Himalayan tahr, the mountain hare, the yak, the grey fox and the black bear (Fig. 3.4).

DON'T MISS OUT

'Ganga' is the Indian name of the largest river originating in the Himalayas. In English, 'Ganges' is also used. Nearly 2,500 km long, this river has numerous tributaries (that is, other rivers joining it). Some of them, like the Yamuna and the Ghagara, also originate in the Himalayas. Others, like the Son or Sone, originate from the Vindhya Range to the south of the Ganga plain.

Fig. 3.5. Terrace farming in north India

Terrain: A piece or stretch of land, from the point of view of its physical features.

Life in the mountains

The mountain **terrain** is usually rugged or rough, and has steep slopes. This means that regular farming can only be practised in some **valleys**. Cultivation is practised on the

Reprint 2025-26

slopes by cutting steps into the slope (Fig. 3.5). This is called terrace farming. In many mountainous regions of the world, herding is the preferred occupation over agriculture.

Tourism is often an important source of income for the people living in the mountains. The crisp mountain air and scenic beauty attract many tourists. Some tourists also go to the mountains for sports such as skiing, hiking, mountaineering and paragliding. For many centuries, people have also travelled to these uplands for pilgrimages to holy sites. But an excessive inflow of visitors can also put the fragile mountain environment under pressure; it is often difficult to find the right balance.

Valley: A lower area between hills or mountains, often with a river or stream flowing through it.

DON'T MISS OUT

- **Bachendri Pal** started climbing mountains from a young age and led many women's climbing expeditions. She was the first Indian woman to climb Mount Everest in 1984 and was awarded Padma Shri the same year (and Padma Bhushan in 2019).
- **Arunima Sinha** lost a leg in an accident when she was 22. With Bachendri Pal's encouragement and training, she managed to climb Mount Everest in 2013, and went on to climb the highest peak of every continent, including Mount Vinson in Antarctica! She was awarded Padma Shri in 2015.

LET'S EXPLORE

These images (Fig. 3.6 on page 50) depict a few challenges that people living in the mountains may face. Discuss them in groups in the class and write one paragraph on each. Also discuss why, despite many such challenges, people still choose to live in the mountains.

Reprint 2025-26

Flash flood: A sudden local flood, often caused by a cloudburst.

Landslide: The sudden collapse of a mass of earth or rock from a mountainside.

Avalanche: The sudden fall of snow, ice or rocks from a mountainside; often occurs when the snow starts melting.

Cloudburst: A sudden violent rainstorm.

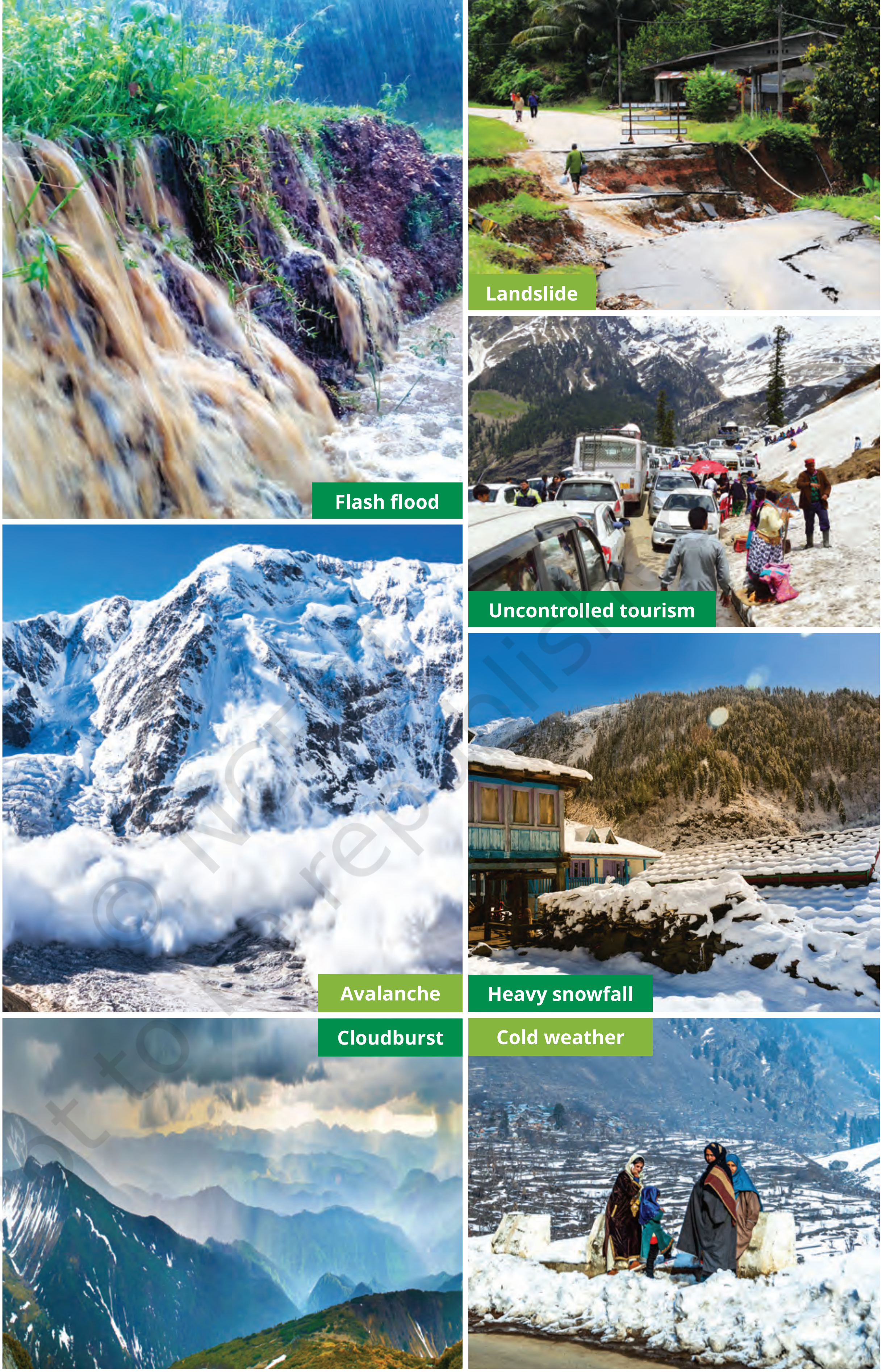


Fig. 3.6. Life in the mountains has definite positives, from pure air to the beautiful scenery. It also involves potential challenges, both natural and human-made, some of which are depicted in these pictures.

DON'T MISS OUT

Many traditional communities around the world consider mountains to be sacred places and worship them. Mount Everest, the highest mountain in the world at 8,849 m, has many names. Tibetans call it 'Chomolungma', which means 'Mother Goddess of the World' and worship the mountain as such. Nepalis call it 'Sagarmatha', meaning 'Goddess of the Sky'. Similarly, Mount Kailash in Tibet is held sacred by followers of Hinduism, Buddhism, Jainism and Bon (an ancient Tibetan religion). Such reverence for mountain summits is also found elsewhere in India, as well as in other parts of the world.

Plateaus

A plateau is a landform that rises up from the surrounding land and has a more or less flat surface; some of its sides are often steep slopes. Like mountains, plateaus can be young or old in terms of the Earth's history. Two examples of plateaus are the Tibetan Plateau, the largest and highest plateau in the world, and the Deccan Plateau. The height of plateaus can vary from a few hundred metres to several thousand metres.

DON'T MISS OUT

- The Tibetan Plateau has an average altitude of 4,500 m, which explains why it has been nicknamed the 'Roof of the World'! From east to west, it is nearly 2,500 km long — the distance from Chandigarh to Kanyakumari.
- The Deccan Plateau of central and south India is one of the oldest plateaus in the world, formed through volcanic activity millions of years ago.

Like mountains, plateaus are rich in mineral deposits; they have been called 'storehouses of minerals'. As a result, mining is a major activity on plateaus, where many of the world's largest mines are found. For example, the East

Reprint 2025-26

African Plateau is famous for gold and diamond mining. In India, huge reserves of iron, coal and manganese are found in the Chhota Nagpur Plateau.

The plateau environment is very diverse across the world. Many plateaus have a rocky soil, which makes them less fertile than plains (see next section) and therefore less favourable to farming. An exception is that of lava plateaus (that is, formed through volcanic activity), as they often have a rich black soil.

Plateaus are also home to many spectacular waterfalls. The Victoria Falls on the Zambezi River in southern Africa, the Hundru Falls on the Subarnarekha River in the Chhota Nagpur Plateau and the Jog Falls on the Sharavati River in the Western Ghats are a few such waterfalls. The Nohkalikai Falls (Fig. 3.7) drop down 340 metres from the Cherrapunji Plateau (in Meghalaya).

Fig. 3.7. The Nohkalikai Falls emerging from the Cherrapunji Plateau

Reprint 2025-26

Plains

Plains are landforms that have an extensive flat or gently undulating surface. They do not have any large hills or deep valleys. They are generally not more than 300 metres above **sea level**.

Floodplains are one type of plains formed by rivers originating in mountain ranges, where they collect particles of rock, sand and silt called 'sediments'. These sediments are carried all the way to the plains, where the rivers deposit them, making the soil very fertile. As a result, these plains are ideal for growing crops of all kinds, and agriculture is a major economic occupation in this landform. Plains also support a variety of flora and fauna.

Sea level: The average level of the surface of the oceans, also called 'mean sea level'.

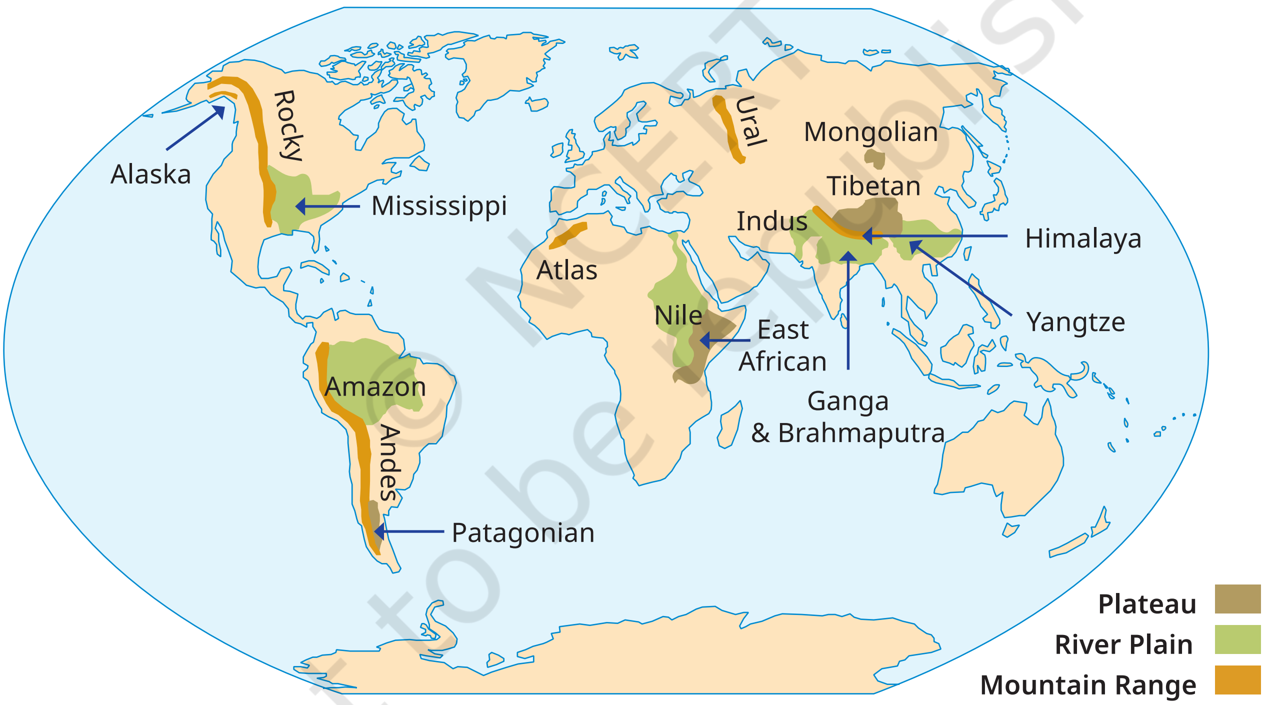


Fig. 3.8. This world map shows a few major mountain ranges, plateaus and plains.

LET'S EXPLORE

Use the colour code in Fig. 3.8 to add a landform to each name. For instance, 'Tibetan plateau', 'Rocky range', 'Nile plain'. *(You do not have to remember the names in this map.)*

Reprint 2025-26

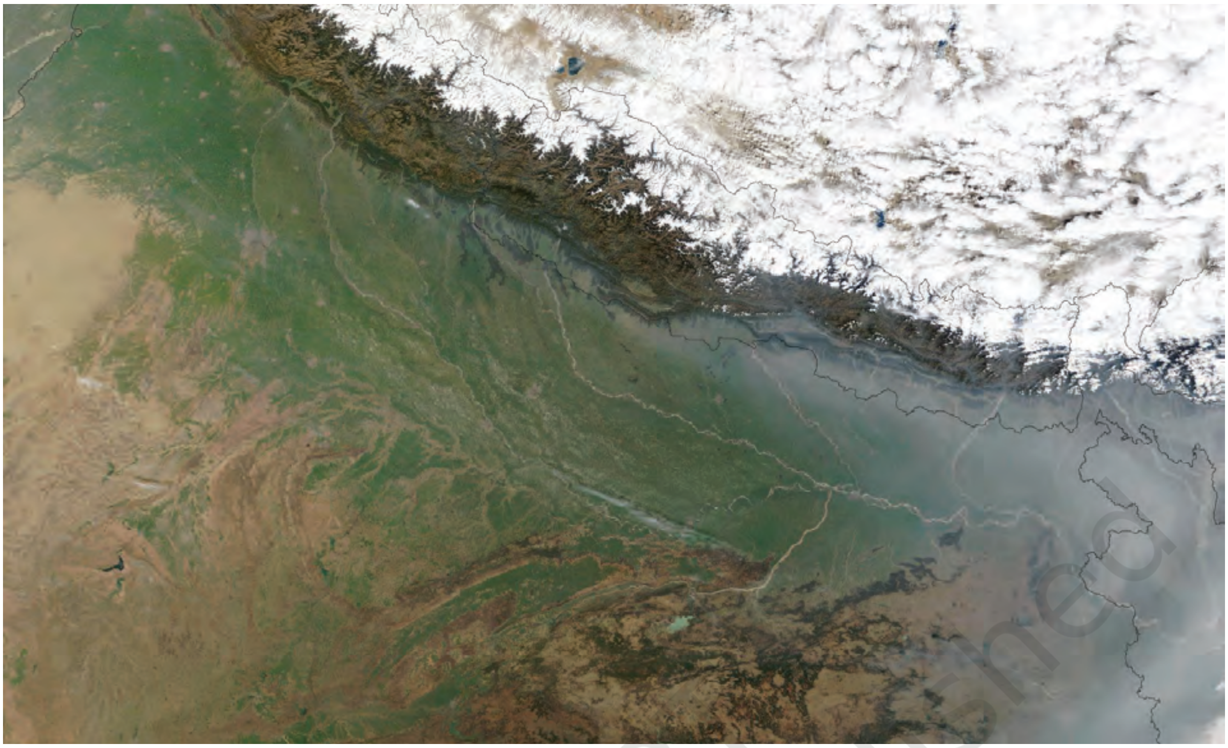


Fig. 3.9. A satellite view of the Ganga plain

LET'S EXPLORE

The picture in Fig. 3.9 has been taken from a satellite. It captures a portion of north India from a high altitude. Observe and discuss the image as a class activity.

- → Which colour is the Ganga plain?
- → What does the white expanse represent?
- → What does the brown expanse at the bottom left of the image represent?

Life in the plains

Thousands of years ago, the first civilisations developed around rivers in fertile plains. In our times too, a large part of the world's population lives in plains.

About 40 crore people, more than one-fourth of the total Indian population, live in India's Ganga plain (often called the 'Gangetic plain'). As with many other plains of the world, the major occupations of people in this region include river fishing and agriculture. Food crops such as rice, wheat, maize, barley and millets are grown. Fibre crops such as cotton, jute and hemp are also grown in the Gangetic plain. Traditional agriculture has been mostly rainfed (that is, watered through rainfall). In recent decades, however, agriculture has turned to irrigation, with water brought to the fields through networks of canals or pumped from groundwater. While irrigation has increased agricultural production, it has also contributed to the depletion (or decrease) of groundwater. This presents a challenge for the future of agriculture in the region. Some of the other problems affecting the Ganga plains include high population and pollution.

Whether in mountain ranges or plains, rivers around the world have carried immense cultural value. In particular, many communities have considered a river's source and its **confluence** with one or two other rivers to be sacred. In India, numerous festivals, ceremonies and rituals are conducted at such locations.

Because plains have a gentle slope, river navigation is easy and supports a lot of economic activities. In earlier days, people also used rivers extensively to travel from one place to another. Even today (Fig. 3.10 on page 56), there are stretches along the Ganga where people prefer to use boats to move around!

Confluence: The meeting point of two or more rivers.

LET'S EXPLORE

- Can you give examples of river sources or confluences from your region that are regarded sacred by any community?
- Visit a nearby river and observe all activities there, whether economic or cultural. Note them down and discuss with your classmates.

Reprint 2025-26

→ Name some popular tourist destinations in India and identify the category of landform they are associated with.

Fig. 3.10. River transportation on the Ganga

Resilience: The capacity to meet challenges and difficulties, adapt to them or overcome them.

In this chapter, we explored the three main landforms. But its surface is very complex and experts often define a few more landforms. One such landform is the desert. Deserts are considered to be large and dry expanses with very little precipitation. Their flora and fauna are also unique. Some deserts are hot, like the Sahara Desert in Africa or the Thar Desert in the northwest of the Indian Subcontinent. Others are cold, like the Gobi Desert in Asia. (Some experts also describe the Antarctica continent as a desert.)

Despite harsh living conditions, humans have adapted to most of the deserts. In India, communities living in the Thar Desert, or migrating through it, hold rich cultural traditions, such as folk songs and legends, related to the desert.

The diverse ways in which humans have made all landforms their home is a testimony to our adaptability and **resilience**.

Reprint 2025-26

The five *tiṇais* of ancient Tamil Sangam poetry are five landscapes associated with certain specific deities, lifestyles, moods or emotions (such as love, longing, separation, quarrel, etc.). This table only lists the characteristics of the five landscapes and the main human occupations in each:

Tiṇai	Landscape	Main occupation
Kuriñji	mountainous regions	hunting and gathering
Mullai	grassland and forests	cattle rearing
Marudam	fertile agricultural plains	farming
Neydal	coastal regions	fishing and seafaring
Pālai	arid, desert-like regions	journeying and fighting

These five *tiṇais* constitute a different classification of landforms than the one we have seen, but they reflect a keen awareness of the diverse regions and their characteristics. They also illustrate the deep connection between humanity and the natural environment. *(You do not need to remember the details of the tiṇais, but the concepts they reflect need to be understood.)*

Before we move on …

- Landforms are classified into three main types — mountains, plateaus and plains. They have very different physical characteristics and environments.
- Throughout history, people's lives and activities have been much impacted by the type of landform they have lived in. These landforms are an integral part of culture across the world. Indian culture, in particular, has celebrated them in diverse ways.
- Each landform offers different challenges as well as opportunities.

Reprint 2025-26

Questions, activities and projects

1. In what type of landform is your town / village / city located? Which features mentioned in this chapter do you see around you?
2. Let us go back to our initial trip from Chhota Nagpur to Prayagraj and Almora. Describe the three landforms you came across on the way.
3. List a few famous pilgrimage spots in India along with the landforms in which they are found.
4. State whether true or false —
   - The Himalayas are young mountains with rounded tops.
   - Plateaus usually rise sharply at least on one side.
   - Mountains and hills belong to the same type of landform.
   - Mountains, plateaus and rivers in India have the same types of flora and fauna.
   - Ganga is a tributary to the Yamuna.
   - Deserts have unique flora and fauna.
   - Melting snow feeds rivers.
   - Sediments from rivers deposited in the plains makes the land fertile.
   - All deserts are hot.
5. Match words in pairs:

Mount Everest	Africa
rafting	roof of the world
camels	rice fields
plateau	desert
Gangetic plains	river
waterway	Ganga
Mount Kilimanjaro	tributary
Yamuna	climbing

Reprint 2025-26

CHAPTER 8

Unity in Diversity, or 'Many in the One'

Oh, grant me my prayer, that I may never lose the bliss of the touch of the one in the play of the many.

— Rabindranath Tagore

... The principle of unity in diversity which has always been normal to [India] and its fulfilment the fundamental course of her being and its very nature, the Many in the One, would place her on the sure foundation of her Swabhava and Swadharma.

— Sri Aurobindo

The Big Questions ?

1. *What is meant by 'unity in diversity' in the Indian scenario?*
2. *What aspects of India's diversity are the most striking?*
3. *How do we make out the unity underlying the diversity?*

0681CH08

A Rich Diversity

If you travel through India by train, you will notice not only changing landscapes but also many different types of dresses and food; you will hear different languages, familiar and unfamiliar, and see different scripts on the way. Even within your own region, you will often come across people from other parts of India with different customs and traditions. This is India's rich diversity, and it is usually the first thing that strikes visitors to our country.

With over 1.4 billion inhabitants (about 18 percent of the world's population), such diversity is not surprising! In the late 20th century, the Anthropological Survey of India, a national organisation, conducted a massive survey called 'People of India project' of 4,635 communities across all States of the country. It counted 325 languages using 25 scripts; it also observed that many Indians may be called migrants, in the sense of people not living near their birthplace or with their original community.

LET'S EXPLORE

As a class activity, make lists of (1) the birthplaces of at least 5 classmates and the birthplaces of their parents; (2) the students' mother tongues and other languages known to them. Discuss the results in terms of diversity.

While diversity is indeed beautiful, making sense of it is not so easy. Over a century ago, the British historian Vincent Smith wondered,

> "How, in the face of such bewildering diversity, can a history of India be written? … The answer to the query is found in the fact that India offers **unity in diversity**."

What is meant by 'unity in diversity'? How shall we perceive and express this unity, or the 'Many in the One'? To answer this question, we will explore a few dimensions of Indian life.

Reprint 2025-26

Food for All

Some of you will have eaten food from different regions of India. The number of different dishes and preparations you can taste in India must be in their thousands, if not lakhs! Yet certain food grains are common to almost every part of the country — cereals such as rice, barley and wheat; millets such as pearl millet (bajra), sorghum (jowar), finger millet (ragi); and pulses such as various kinds of dals and

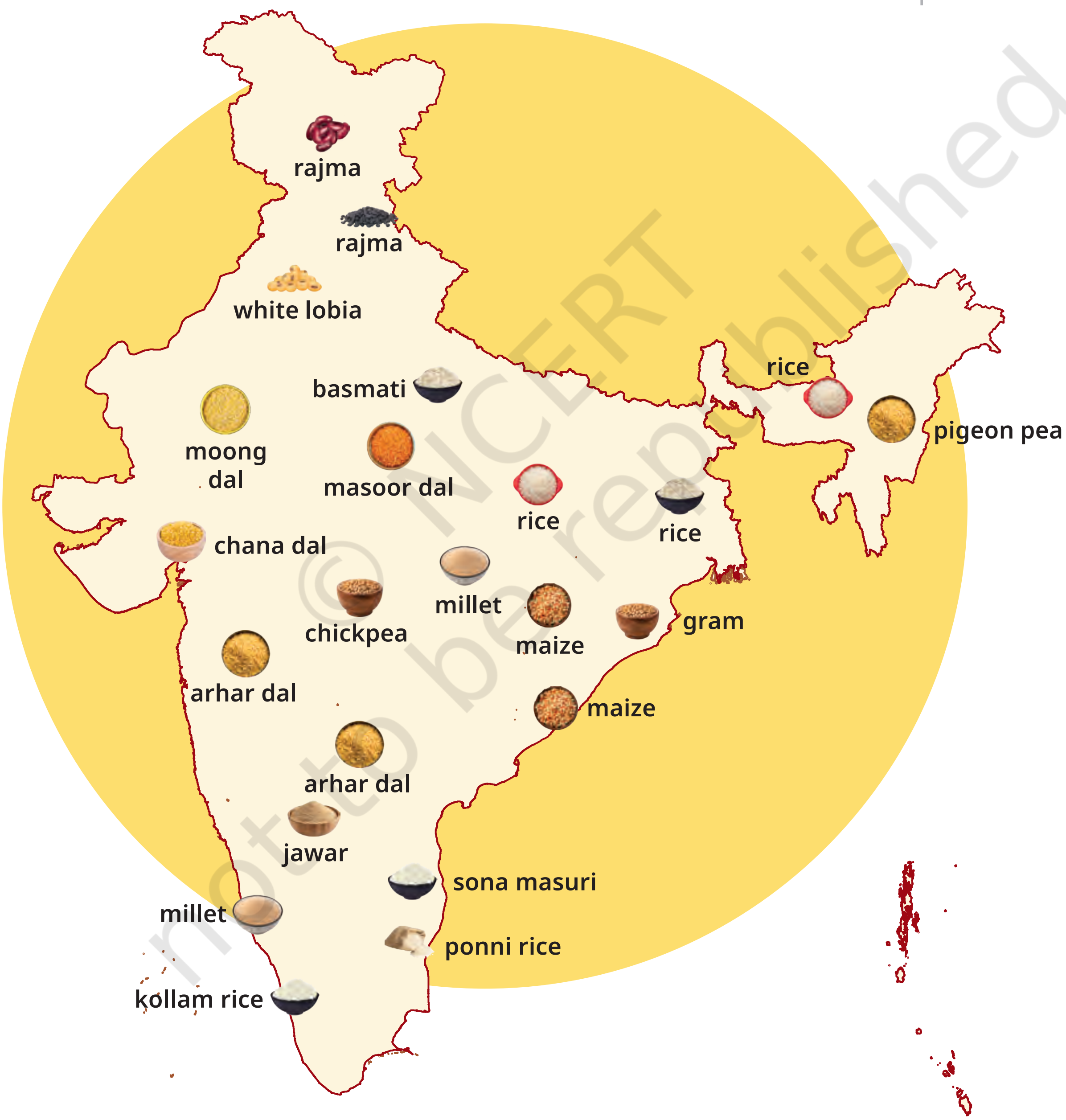


Fig. 8.1. A few examples of cereals and pulses from different regions of India

grams. All these are called 'staple grains' because they are the basic food for most Indians (Fig. 8.1 on page 127). Similarly, some common spices such as turmeric, cumin, cardamom and ginger, are also used throughout the country. We could continue this list with some common vegetables, common oils etc.

So we see how the same ingredients *(unity)* can be used in a number of combinations *(diversity)* to prepare an endless variety of dishes!

LET'S EXPLORE

→ As a class activity, make a list of ingredients (grains, spices, etc.) that are used in your home.

→ Take any one vegetable and think of the number of different dishes you can prepare with it.

Textiles and Clothing

Fig. 8.2. Stone relief of a woman in a sari from Vaiśhali (today in Bihar)

Every region and community in India has developed its own styles of clothing and dresses. Yet, we notice a commonality in some traditional Indian dresses, irrespective of the material used. An obvious example is the plain length of cloth called the sari, a type of clothing worn in most parts of India and made from different fabrics — mostly cotton or silk, but nowadays synthetic fabrics too. Banarasi, Kanjivaram, Paithani, Patan Patola, Muga or Mysore are some of the famous types of silk saris. There are many more kinds of cotton saris. Altogether, this unstitched piece of cloth comes in hundreds of varieties. They are

Reprint 2025-26

produced by different methods of weaving (Fig. 8.3 on the right) and designing. Some designs are part of the cloth, while others are printed after the cloth is woven. Finally, there are endless variations in the colours, which are produced from many kinds of pigments.

The sari has a long history. This stone **relief** (Fig. 8.2 on page 128) from Vaiśhali (today in Bihar) goes back a few centuries BCE.

Fig. 8.3. A few specimens of colourful traditional Indian textiles.

LET'S EXPLORE

Explain how the example of the sari reflects both unity and diversity (in 100-150 words).

DON'T MISS OUT

For a very long time, India produced the finest cotton in the world and Indian textiles were exported as far away as Europe. One beautiful type of printed cotton called 'chintz' became so popular in 17th-century Europe that the sale of some European dresses dropped sharply. Eventually, to protect their own products, England and France decided to ban the import of chintz from India!

Relief: A design that stands out from the surface of a panel (which may be of stone, wood, ceramic or another material).

There are many ways of wearing the sari, as they vary from one region to another or from one community to another. In fact, new ways of draping it are still being invented. But in the end, it is a single dress — the sari. In past centuries,

Reprint 2025-26

Fig. 8.4. Women often put the sari to many uses beyond that of a dress (pictures from south India).

several travellers to India marvelled at its simplicity, economy, and the diverse ways in which it is worn. In addition, women often put the sari to many uses beyond that of a dress; the six pictures in Fig. 8.4 above illustrate a few such creative uses.

LET'S EXPLORE

→ In the above pictures, can you recognise what a sari has been used for?

→ Are you aware of, or can you imagine, more uses for the sari?

Reprint 2025-26

→ Following the example of the sari, make a list of different styles you have seen for the dhoti — both as regards the fabric and the uses the dhoti can be put to. What conclusion can you draw?

Festivals Galore

There is an immense variety of festivals in India. You may have noticed that a few common ones are celebrated across India almost at the same time, though they have different names. We will take just one example — Makara Sankrānti, which marks the beginning of the harvest season in many

Reprint 2025-26

parts of India on or around January 14. The map shows the different names of similar festivals across India about the same date.

Fig. 8.5. Different names of similar festivals across India about the same date

LET'S EXPLORE

→ What is your favourite festival and how is it celebrated in your region? Do you know whether it is celebrated in any other part of India, maybe under a different name?

Reprint 2025-26

→ During October–November, many major festivals take place in India. Make a list of the few main ones and their various names in different parts of the country.

An Epic Spread

Literature offers us another fine illustration of unity in diversity. Indian literatures are extremely diverse (and among the most abundant in the world). Over centuries, despite differences in language, technique, etc., they have shared important themes and concerns. Who has not heard of the *Pañchatantra*, for example? This collection of delightful stories, with animals as the main characters, teaches us important life skills. The original Sanskrit text is at least 2,200 years old, but its stories have been adapted in almost every Indian language. In fact, they have travelled well beyond India, all the way to Southeast Asia, the Arab world and Europe, inspiring new collections of stories on the way — it is estimated that about 200 adaptations of the *Pañchatantra* exist in more than 50 languages! This illustrates how 'one' collection of stories has become 'many'.

The most striking case, however, is that of India's two **epics** — the Rāmāyaṇa and the Mahābhārata.

Epic: A long poem generally narrating the adventures of heroes and other great figures of the past.

These two long Sanskrit poems, which together might fill some 7,000 pages in their original versions, narrate the stories of heroes who fight to re-establish dharma. In the Mahābhārata, the Pāṇḍavas, with Kṛiṣhṇa's help, fight their own cousins, the Kauravas, to recover their kingdom. In the Rāmāyaṇa, Rāma, with the help of his brother Lakṣhmaṇa and of Hanuman, defeats the demon Rāvaṇa, who had kidnapped his wife Sītā. These stories contain many shorter ones that focus on values, and constantly ask questions about what is right and what is wrong.

Reprint 2025-26

Fig. 8.6. A painting depicting a major episode from the Rāmāyaṇa (18th century, Himachal Pradesh)

LET'S EXPLORE

In a class discussion, try to identify the episode depicted in the painting shown in Fig. 8.6 above and important details associated with it.

For more than two millenniums, these two epics have been translated or adapted into regional literatures in India and beyond. In addition, there are countless folk versions of them. A few years ago, a scholar conducted a survey in Tamil Nadu alone and counted "about a hundred versions [of the Mahābhārata] that have come down to us in folklore forms." Imagine what the number might be for the whole of India!

In fact, many communities have their own versions of the Rāmāyaṇa and the Mahābhārata. They have also preserved legends connecting their own history with these epics. This is especially true of tribal communities in many parts of

India, such as the Bhils, the Gonds, the Mundas and many more. Most tribes of India's northeast and Himalayan regions, including Kashmir, have had their own version of one or the other of the two epics, or both. These tribal adaptations are transmitted orally, along with legends on how the heroes of the Rāmāyaṇa and the Mahābhārata (generally any or all of the five Pāṇḍavas, their wife Draupadī, but also sometimes their cousin and adversary Duryodhana) visited the tribes' respective regions.

Fig. 8.7. 'Pañcha Pāṇḍavar', a carved stone in a forest of the Nilgiris, Tamil Nadu, depicting the five Pāṇḍava brothers. The shrine containing this stone is maintained by Irula tribals to commemorate the Pāṇḍavas' passing through the area.

The anthropologist K.S. Singh directed the 'People of India' project we referred to earlier (see page 126). In the case of the Mahābhārata, he observed, "There is hardly a place in the country which the epic heroes such as the Pandavas, did not visit according to folklores." And the same may be said of the Rāmāyaṇa. Over the centuries, perhaps more than any other texts, these two epics created a dense web of cultural interactions across India and many parts of Asia. Another example of unity in diversity.

To further illustrate the theme of this chapter, we could have continued our journey and turned to more facets of Indian culture. For instance, in India's classical arts, including classical architecture, both diversity and unity are easily noticeable. (You will study these fields through the Art Curricular Area.)

In the end, we should remember that Indian culture celebrates diversity as an enrichment, but never loses sight of the underlying unity which nourishes that diversity.

Reprint 2025-26

Before we move on …

- India offers immense diversity in its landscapes, people, languages, dresses, foods, festivals and customs.
- Diversity is easy to perceive in many fields, but there is also an underlying unity.
- India's unity celebrates diversity because diversity does not divide — it enriches.

Questions, activities and projects

1. Conduct a class discussion on the two quotations at the start of the chapter.
2. Select a few stories from the *Pañchatantra* and discuss how their message is still valid today. Do you know of any similar stories from your region?
3. Collect a few folk tales from your region and discuss their message.
4. Is there any ancient story that you have seen being depicted through a form of art? It could be a sculpture, a painting, a dance performance, a movie … Discuss with your classmates.
5. Discuss in class the following quotation by India's first prime minister, Jawaharlal Nehru, when he travelled to many parts of India before Independence:

> "Everywhere I found a cultural background which had exerted a powerful influence on their lives. … The old epics of India, the Ramayana and the Mahabharata and other books, in popular translations and paraphrases, were widely known among the masses, and every incident and story and moral in them was engraved on the popular mind and gave a richness and content to it. Illiterate villagers would know hundreds of verses by heart and their conversation would be full of references to them or to some story with a moral, enshrined in some old classic."

Reprint 2025-26